

Docking using Maestro: Step-by-step guide

Structure-Based Virtual Screening - an important early stage of drug discovery in which protein structures and molecular modeling are utilized to identify molecules that can be further developed into drugs.

Goal: learn how to perform a virtual screen for **potential inhibitors of FXa** using the ligand docking application Glide (part of Maestro).

Learning objectives:

- Learn the steps of a molecular docking workflow using Schrödinger's Glide.
- Perform a structure-based virtual screen of a small set of ligands
- Design your own inhibitor for FXa and determine its docking score

You will learn how to:

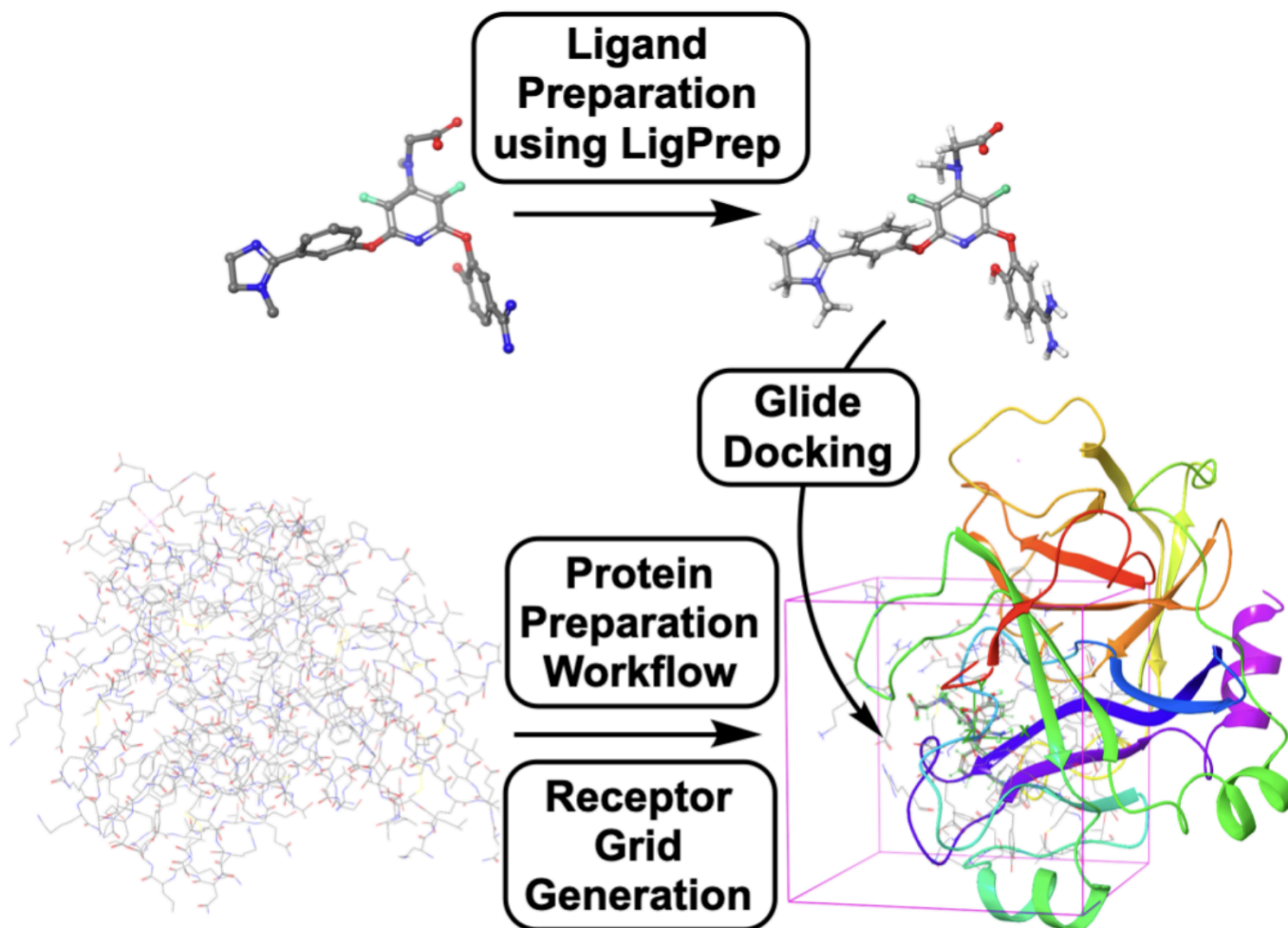
- generate a protein receptor grid,
- dock a set of ligands into the receptor grid,
- analyze the docking results.

You'll then sketch your own ligand designs and evaluate its value.

Independent work with literature

Read and analyse following article <https://www.frontiersin.org/articles/10.3389/fchem.2020.00343/full> and find answers on following questions:

- 1) What is a typical drug development timeline from drug target identification to clinical trials?
- 2) What are some advantages and disadvantages to performing structure-based virtual screens?



Glide is a Schrödinger module that performs ligand-receptor docking reliably. To run a Glide virtual screen, you need a grid file and a ligand file. The grid file is typically generated from a prepared protein (using Protein Preparation Workflow and Receptor Grid Generation), and the ligand file is processed by LigPrep.

Virtual Screening Prerequisites

1.1.Target preparation

Structure files obtained from the PDB and other sources often lack necessary information for performing modeling-related tasks.

Typically, these files are missing hydrogens, partial charges, side chains, and/or whole loop regions. In order to make these structures suitable for modeling tasks you first need to prepare protein (target) to resolve the issues described above.

1.2.Ligand preparation

Similarly, ligand files can be sourced from numerous databases, often in the form of 1D or 2D structures with unstandardized chemistry. LigPrep can convert ligand files to 3D structures, with the chemistry properly standardized and extrapolated, ready for use in virtual screening.

Following steps you need to perform:

- protein preparation,
- cognate ligand preparation,
- virtual screening ligands preparation.

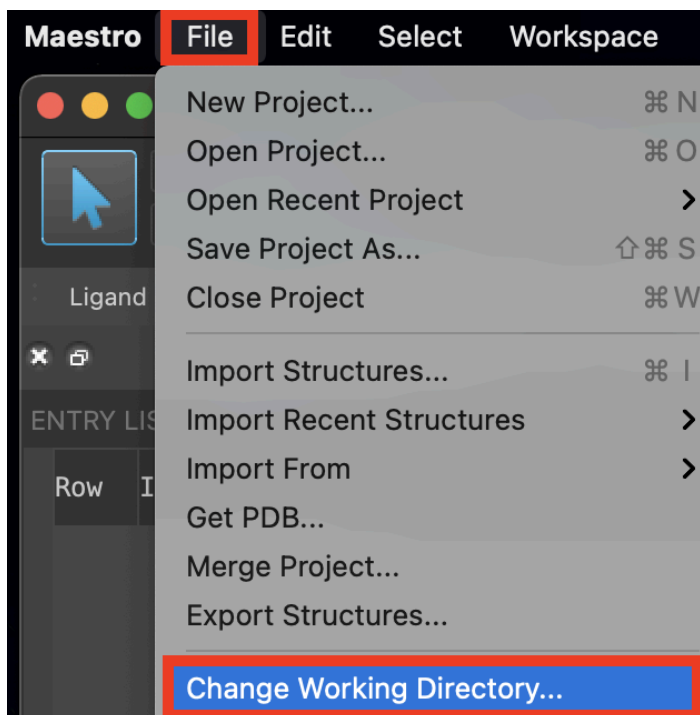
Setup and environment preparation

These preparation steps are a necessary part of a virtual screen and must be done before docking.

During this step we're going to:

- Open Maestro and loading Schrödinger modules
- Set up working directory
- Create and save a project (FXa_Glide.prj)
- Import structures (from PDB or local files)

Open Maestro	in terminal: 1) check available modules: `module avail` 2) find within provided modules following: ... 3) load this module so you can use it: `module load schrodinger/2024-1` (or just most recent version of schrodinger/[year]-[month]) 4) run maestro: `maestro`
	Window > Project Table - Now you have opened Project Table .
<u>Creating Projects and Importing Structures</u>	



Change Working Directory option.

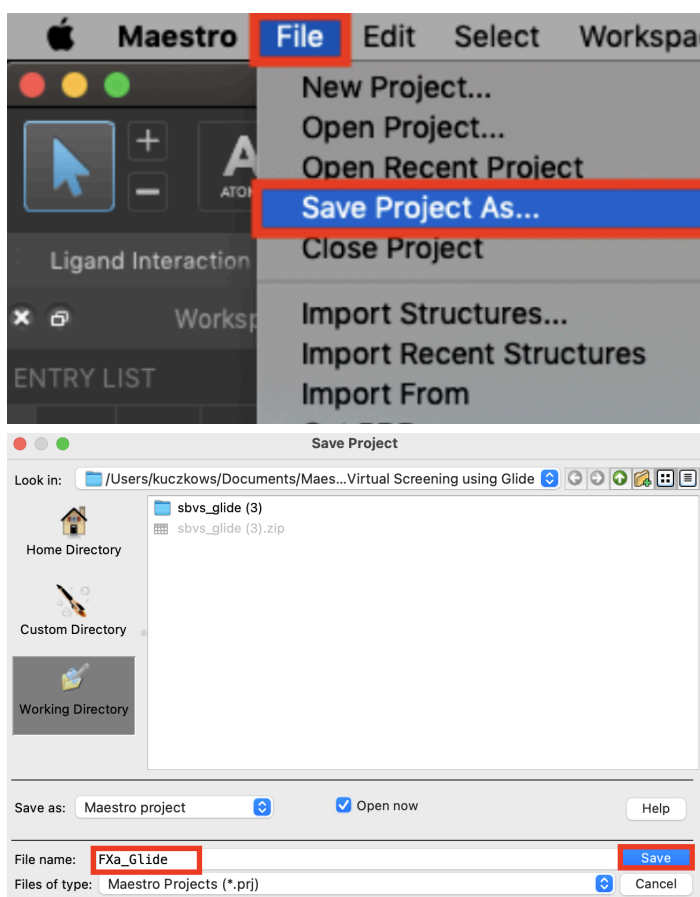
At the start of the session, change the file path to your chosen *Working Directory* in Maestro. That makes file navigation much easier.

Each session in Maestro begins with a default Scratch Project, which is not saved.

Scratch Project - a temporary project in which work is not saved, closing a scratch project removes all current work and begins a new scratch project.

This is why it is important to create a new project at the beginning of the work in your *Working Directory*. In this case all the changes will be saved automatically.

Go to **File > Change Working Directory**
Find your working folder that you created for saving docking results, and click **Choose**.



Save Project panel.

A Maestro project stores all your data and has a .prj extension. A project may contain numerous entries corresponding to imported structures, as well as the output of modeling-related tasks. Once a project is created, the project is automatically saved each time a change is made.

Go to **File > Save Project As...**
Type name of your project in format "name_of_your_project" and click **Save**
The project will be titled name_of_your_project.prj

In our case it would be better to name the project based on the name of the target protein we're working with: **FXa_Glide.prj**.

Now our Maestro project has the name: FXa_Glide.prj.

Structures can be imported:

- from the PDB directly
- from your *Working Directory* using **File > Import Structures**, and are added to the Entry List and Project Table.

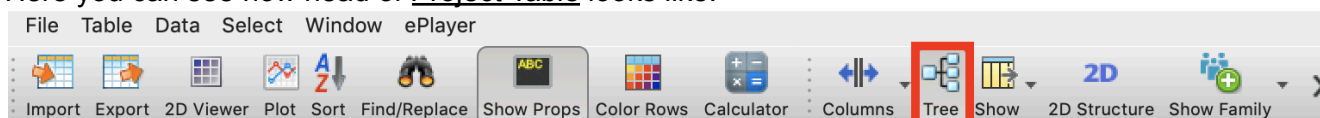
Difference between Entry List and Project Table?

- Entry List - a simplified view of the Project Table that allows you to perform basic operations such as selection and inclusion

- The Entry List is located to the left of the Workspace. Workspace - the 3D display area in the center of the main window, where molecular structures are displayed.
- Here you can see how Entry List looks like:

Row	In	Title
1	☐	1fjs_prep_lig
2	☐	1fjs_prep_receptor
3	☐	1fjs_prep_complex
		▶ 50ligs_epik (92)
		▶ ligprep_1FJS-out1 (3)
		▶ glide_1FJS-cognate_pv1 (4)
	2	glide_1FJS_screening_pv1 (2...
103	☒	1fjs_prep_complex
104	☒	15650 Known Active
105	☐	15722 Known Active
106	☐	1167 Known Active
107	☐	16088 Known Active
108	☐	15722 Known Active
109	☐	15722 Known Active

- Everything that is highlighted with blue "In" circle within the Entry List is in the Workspace (you can see it on the screen from the right side).
- Project Table - displays the contents of a project and is also an interface for performing operations on selected entries, viewing properties, and organizing structures and data.
 - The Project Table can be accessed by **Ctrl+T (Cmd+T)** or **Window > Project Table** if you would like to see an expanded view of your project data.
 - Here you can see how head of Project Table looks like:



Get PDB File

Note: Downloading will create PDB files in the current directory, and then automatically import them.

PDB IDs:

Chain name (optional):

Include:
 ☐ Diffraction data
 ☐ Biological unit

Fetching from: Local or Web
 Change ▾

Download
Cancel
Help

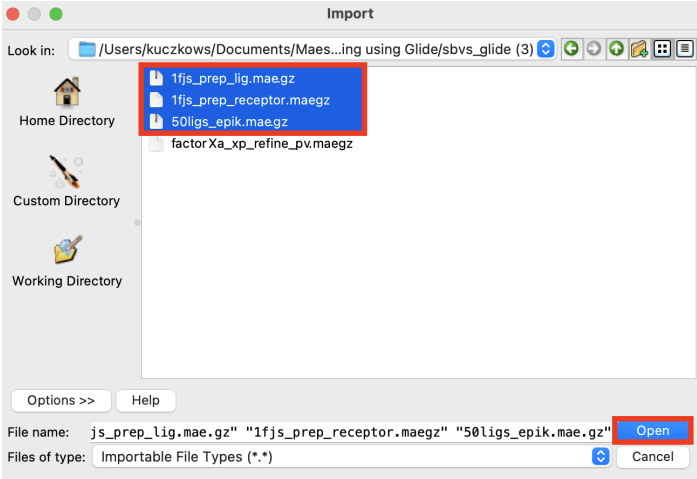
Importing from the PDB directly:

Go to **File > Get PDB**

For PDB IDs, type: **1fjs**

Click **Download**

Result: 1FJS is loaded into the Workspace + banner appears.



The Import panel, with desired files selected.

Importing from your *Working Directory*
Go to **File > Import Structures**

Ctrl-click (Cmd-click) to select files:

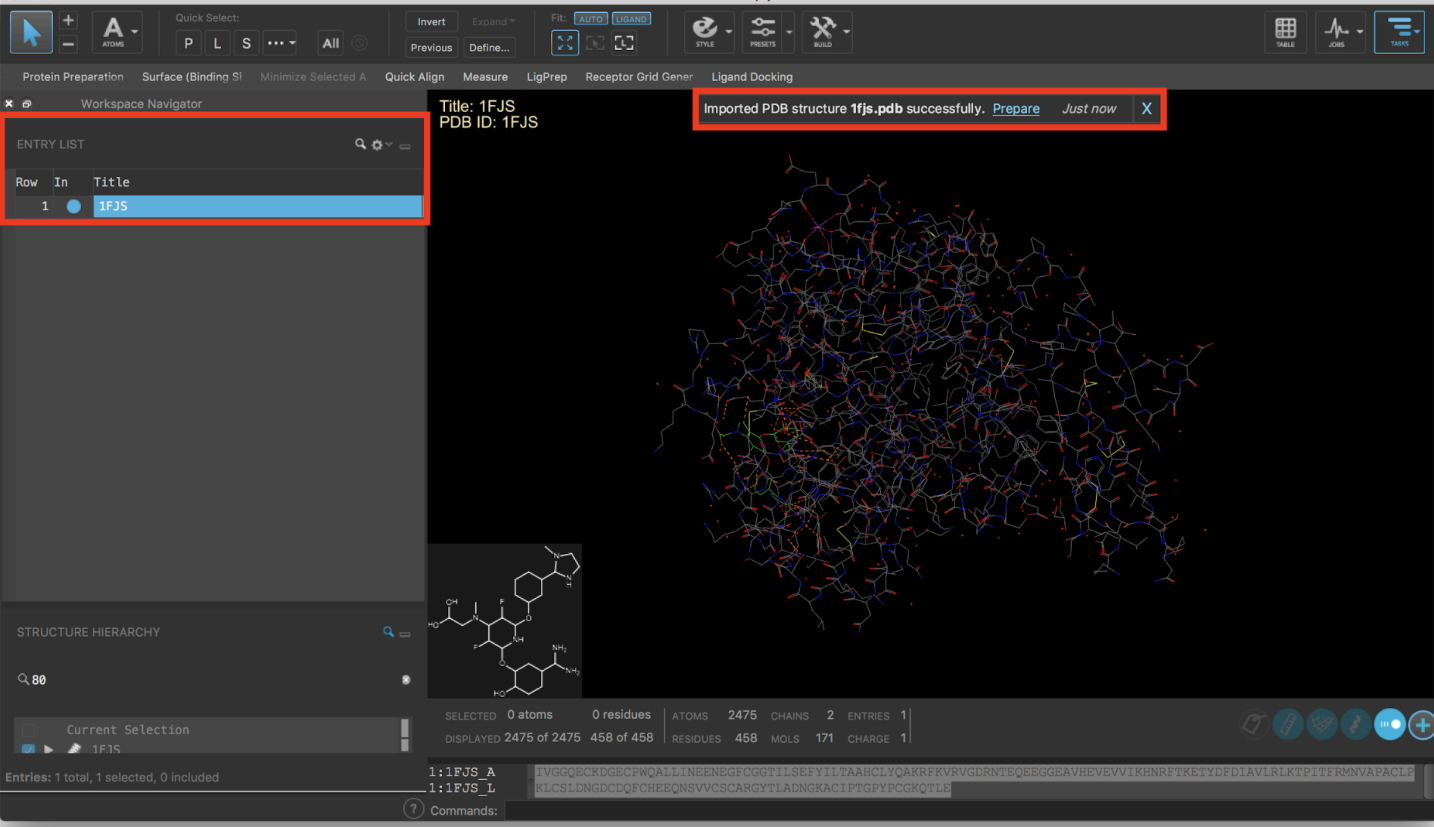
- 1fjs_prep_lig.mae.gz
- 1fjs_prep_receptor.mae.gz
- 50ligs_epik.mae.gz

Click **Open**

- Structures are in the Entry List
- A banner appears confirming entries have been imported

Note: Banners appear when:

- files have been imported
- jobs incorporated:
 - incorporated = once a job is finished, output files from the *Working Directory* are added to the project and shown in the Entry List and Project Table.
- to prompt a common next step.



The Workspace after the structure is imported from the PDB, with the Entry List and banner highlighted.

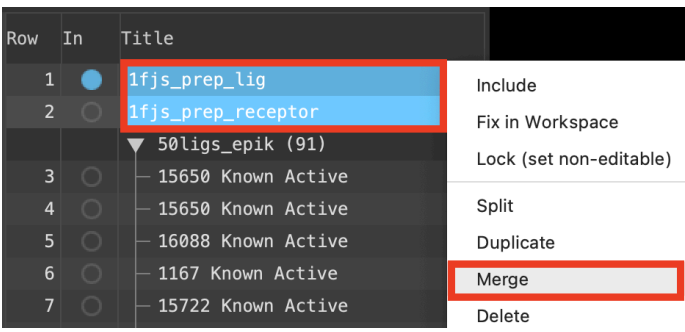
Note: By default the structure corresponding to the imported file is both:

- **included** in the Workspace (the entry is represented in the Workspace, the circle in the “In” column is blue) = included items have filled the blue circle in the “In” column.
- **selected** in the Entry List = selected items are highlighted with blue (blue background; like it presented with red border on image below).

(1) the atoms are chosen in the Workspace. These atoms are referred to as "the selection" or "the atom selection". Workspace operations are performed on the selected atoms.

(2) The entry is chosen in the Entry List (and Project Table) and the row for the entry is highlighted. Project operations are performed on all selected entries)

Row	In	Title
1	☒	1fjs_prep_lig
2	☐	1fjs_prep_receptor
		▼ 50ligs_epik (91)
3	☐	15650 Known Active
4	☐	15650 Known Active
5	☐	16088 Known Active
6	☐	1167 Known Active
7	☐	15722 Known Active

 Merge entries.	Shift-click to select - 1fjs_prep_lig - 1fjs_prep_recep in the <u>Entry List</u> Right-click on the selection and choose Merge Double click on the new entry to rename it 1fjs_prep_complex
--	--

Advice:

Project organisation strategy best practice:

Project_Name/

- ├── input/ (raw PDB files, ligand libraries)
- ├── prepared/ (prepared proteins and ligands)
- ├── grids/ (receptor grids)
- ├── docking/ (docking results)
- ├── analysis/ (analysis outputs, SiteMap results)
- └── figures/ (saved images for publication)

That's not obligatory to follow, but it can make your project structure much easier to understand, especially if it will be extended with time. This organization will make it much easier to track your work progression and find files later.

Protein Structure Preparation

The **Protein Preparation Workflow** is run within the **Preparation Workflow** tab. The workflow has *processing*, *modification*, and *refinement* tools that we will use on our PDB structure. This tool is intended to support two main workflows:

- interactive, single protein preparations. Interactive preparations are manually performed in a step-by-step manner, with the opportunity to review the results of each step and easily control the order of modifications.
- highly automated bulk protein preparations. Automatic preparation is pre-set by the user by the use of toggles that control which stages of the workflow are run in a single job. The Automatic workflow allows processing of multiple protein structures in a single job, permitting they maintain the same settings.

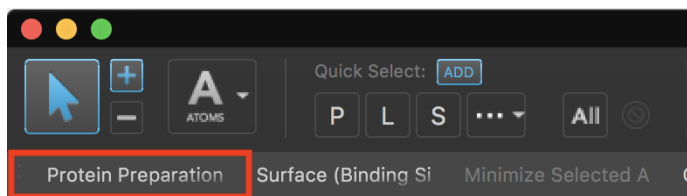
The *recommended minimal processing tasks are checked by default in both workflows* but may be modified using the dropdown options.

There are also options for filling in missing side chains and/or loops, depending on the needs of your structure. For more information see [Protein Preparation Workflow Panel Help](#).

During this step we're going to:

- Understand the Protein Preparation Workflow
- Preprocess the protein (adding hydrogens, fixing bonds)
- Optimize hydrogen bond networks
- Clean up the structure and minimizing
- Review and fix issues (missing side chains, loops)
- Remove unnecessary waters and crystallization artifacts

Preparing Protein Structures for Glide Docking Model

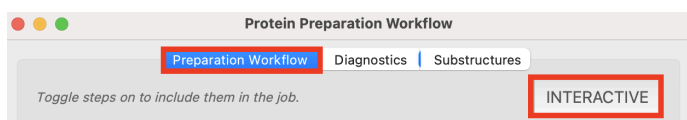


The Protein Preparation Workflow in the Favorites toolbar.

In the **Favorites** toolbar, click **Protein Preparation**. Note: You can also click **Prepare** in the banner or find the **Protein Preparation Workflow** in the **Task Tool**.

Result: **Protein Preparation Workflow** opens.

The **Preparation Workflow** tab may be used in conjunction with the **Diagnostics** tab and **Substructures** tab. These tabs are intended for diagnosis and analysis of the protein after preparation steps because automatic procedures cannot cover all possible cases.



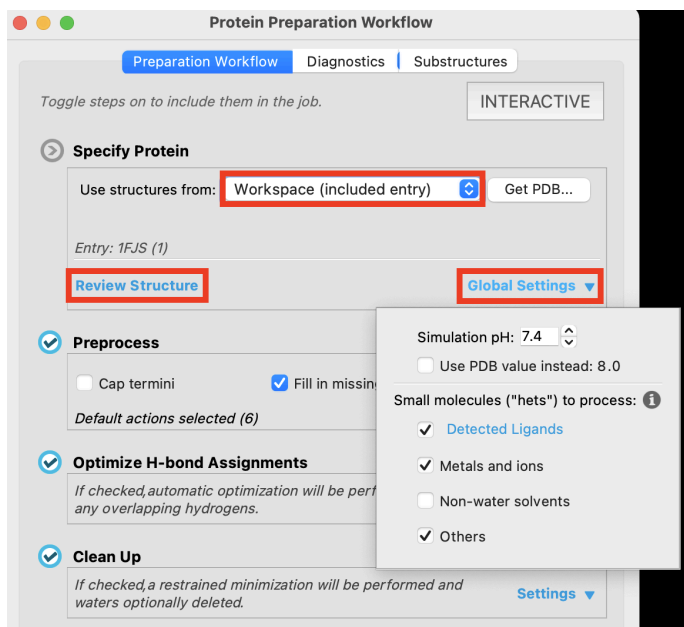
The Protein Preparation Workflow in Automatic mode.

In the **Preparation Workflow** tab, confirm the **INTERACTIVE** button is off (= it is grey, not dark blue).

- NOT INTERACTIVE = fully automated preparation (Maestro will execute all selected operations in sequence without pausing). For automatic mode, you'll see toggle switches for major processing steps.
- INTERACTIVE = a step-by-step interactive approach (it pauses after each operation for user review). For interactive mode, you'll see more granular controls.

To match the preparation workflow to your needs — efficiency for routine preparations or careful control for complex structures.

Note: INTERACTIVE mode can be used for exploring manual options, or to run a single protein in a step-by-step manner. In our case we will be running an automatic protein preparation.



Specify the protein from the Workspace.

In the **Specify Protein** section, choose use structures from: Workspace

Select **Global Settings**:

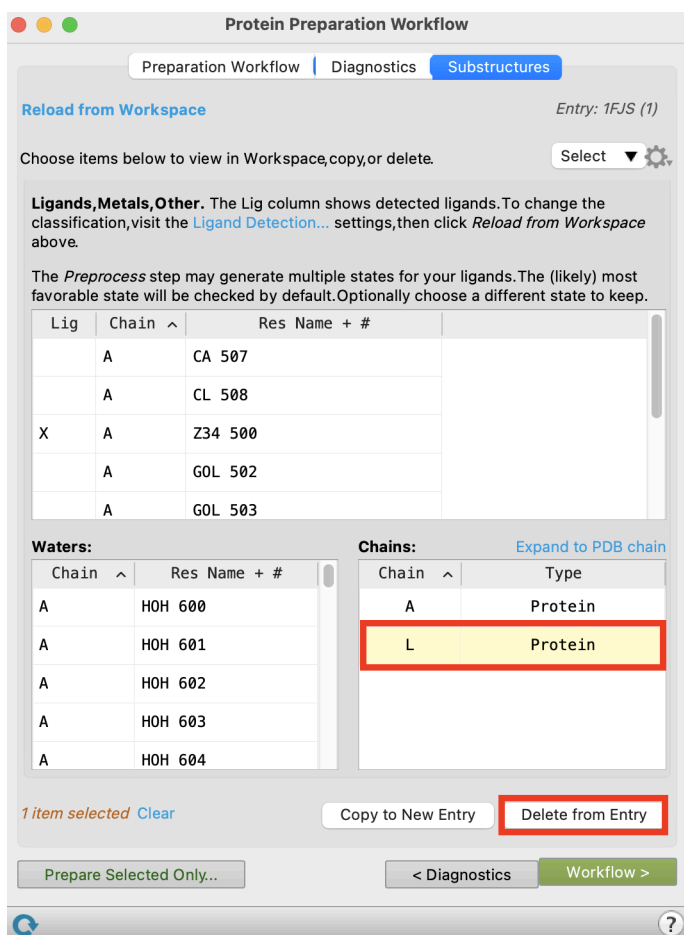
- A dropdown opens showing the simulation pH and the PDB pH as well as small molecule options
- The Non-water solvents option is left unchecked. This saves computational resources because the glycerols in our Workspace will not be prepared

Note: The **Specify Protein** section provides you with the option to prepare a protein from the

- 1) Workspace,
- 2) Project Table,
- 3) File,
- 4) directly from the PDB.

Select **Review Structure**

Result: it will open **Substructures** tab for you to show "Ligands, Metals, Other", "Waters", and "Chains" (just to know)



Review the structure for preparation in the **Substructures** tab.

Maestro categorizes all components of your structure and lets you visualize and selectively delete them.

Let's simplify the structure by removing unnecessary elements (secondary chains) and focus on components that need special attention (relevant protein chain).

Under **Chains**, click Chain L

- The Workspace zooms to the chain

Click **Delete from Entry**

Result: The smaller of the two chains is removed, and a new entry appears in the Entry List ("1FJS - with-deletions").

Note: Unless specified, waters and glycerols (GOL) belonging to chain L will not be removed. Glycerols are a crystallographic artifact with no biological relevance.

The **Select** dropdown provides shortcuts for selecting these species based on their proximity to specified chains. Removing waters and glycerols will be available again in the **Clean Up** and **Analysis** steps (these are future steps, not buttons on the opened "Substructures" panel).

Chain L was removed, creating a new entry called "1FJS - with-deletions".

Protein Preparation Workflow

Preparation Workflow **Diagnostics** Substructures

Check Workspace Entry Entry: 1FJS - with-deletions (2)

One issue was found. See Reports for more information about the protein.

Valences Missing Overlapping Alternates Reports

Valence errors were found. The problem may be missing H or an incorrect number of bonds. The Preprocess step adds H based on the heavy atom. You may also change the element, charge, or number of bonds using the 3D Builder or right-click menu options..

Select table rows to review the corresponding items in the Workspace:

Atom	Residue	Atom type	Expected Bonds	Actual Bonds
N 1	A: ILE 16	NG (35)	4	1
C 2	A: ILE 16	CA (4)	4	3
C 5	A: ILE 16	CA (4)	4	3
C 6	A: ILE 16	CB (5)	4	2
C 7	A: ILE 16	CC (6)	4	1
C 8	A: ILE 16	CC (6)	4	1
N 9	A: VAL 17	NC (29)	3	2
C 10	A: VAL 17	CA (4)	4	3
C 13	A: VAL 17	CA (4)	4	3
C 14	A: VAL 17	CC (6)	4	1
C 15	A: VAL 17	CC (6)	4	1
N 16	A: GLY 18	NC (29)	3	2
C 17	A: GLY 18	CB (5)	4	2

< Workflow Substructures >

Select the **Diagnostics** tab. You will see a comprehensive list of issues categorized by type (Valences, Missing, etc.):

- **Valences:** Valence errors were found, but these bond order issues will be resolved in the protein preparation.
- **Missing:** The Missing tab is empty (it is unavailable to open it), indicating there are no missing side chains in the protein structure.

Select **Reports** to view other issues with the protein that must be resolved prior to modeling.

How does it work? Maestro scans the structure against a database of correct biomolecular structures and flags any deviations.

We need this step to identify all structural defects that could compromise modeling accuracy.

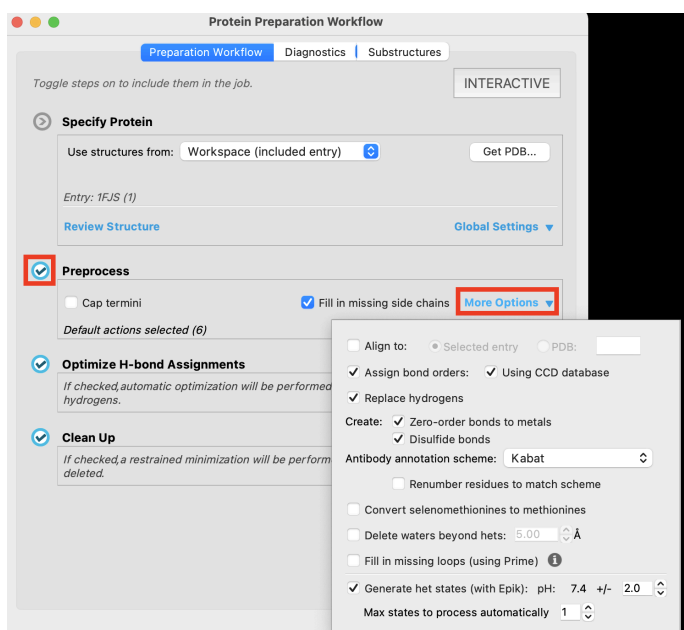
View protein issues in the **Diagnostics** tab

Return to the **Preparation Workflow** tab

The **Protein Preparation Workflow** (“**Preparation Workflow**” tab) toggles include **Preprocess**, **Optimize H-Bond Assignments**, and **Clean up/Minimize and Delete Waters**.

- The **Preprocess** step fixes structural defects and adds missing information.
- The **Optimize H-bond Assignment** section is used for optimizing the hydrogen bonding network – a process that samples water orientations and flips Asn, Gln, and/or His side chains at a specified pH value. Adjusting the pH will change the protonation states of residues and ligands accordingly and is useful if you want to accurately reflect the experimental conditions.
- The **Clean Up** (in some Maestro versions it can be “**Minimize and Delete Waters**”) section fixes clashes that can occur with adding hydrogens or filling missing sidechains. By default, an RMSD of 0.3 Å is used, minimizing both the hydrogens and heavy atoms via harmonic penalty constraints. Optionally, hydrogen-only minimization can be chosen.

Maestro applies chemical rules and databases to convert the often-incomplete PDB structure into a chemically correct molecular model. Maestro performs it to transform the raw crystallographic data into a chemically accurate molecular structure suitable for computational analysis.



Preprocess default settings

Confirm **Preprocess** is toggled on

- Notice Fill in missing side chains is checked by default. If these were missing in your structure they would become populated during this step

Under **Preprocess**, select **More Options**

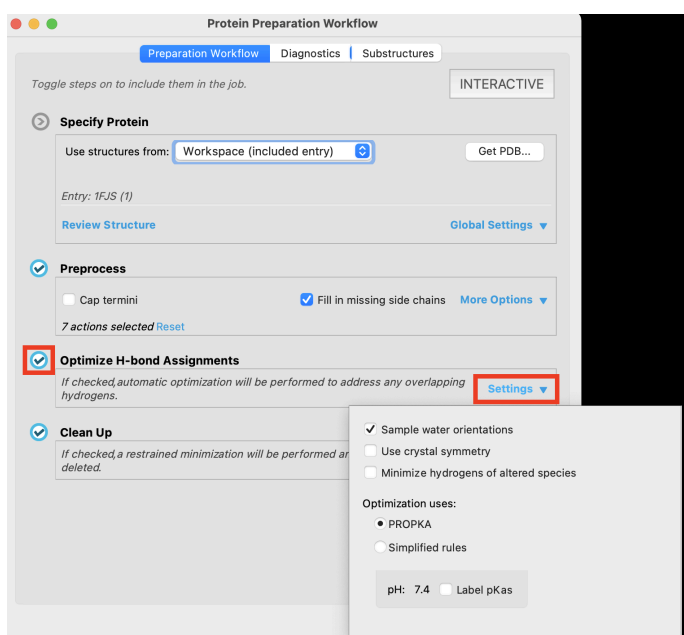
- Among the options provided, notice that missing loops may be filled in using Prime: Fill in missing loops (using Prime). But within our analysis flow we don't need it (don't check it).

Check the pH range for "Generate het states (with Epik)":

- This should align with the physiological or assay pH.

Note: Depending on your system and research question, you may want to keep certain waters. See [Protein Preparation Workflow Panel Help](#) for more details.

The preprocessed structure represents a more realistic molecular model than the raw PDB file. Pay special attention to histidine residues, which can adopt different protonation states, and check that disulfide bonds are correctly formed. The pH setting is particularly important as it determines ionization states of titratable groups, which can significantly impact docking results.



Optimize H-Bond Assignment default settings

Confirm **Optimize H-bond Assignment** is toggled on

What **Optimize H-bond** involves: reorienting hydroxyl hydrogens (Ser, Thr, Tyr), water molecules orientations, and amide groups (Asn, Gln) that may need to be flipped to form the most favorable hydrogen bonding network.

Maestro samples different possible orientations of these groups and selects configurations that maximize hydrogen bonding to resolve ambiguities in hydrogen positioning (not determined by X-ray crystallography) and ensure optimal hydrogen bonding interactions.

Click **Settings**

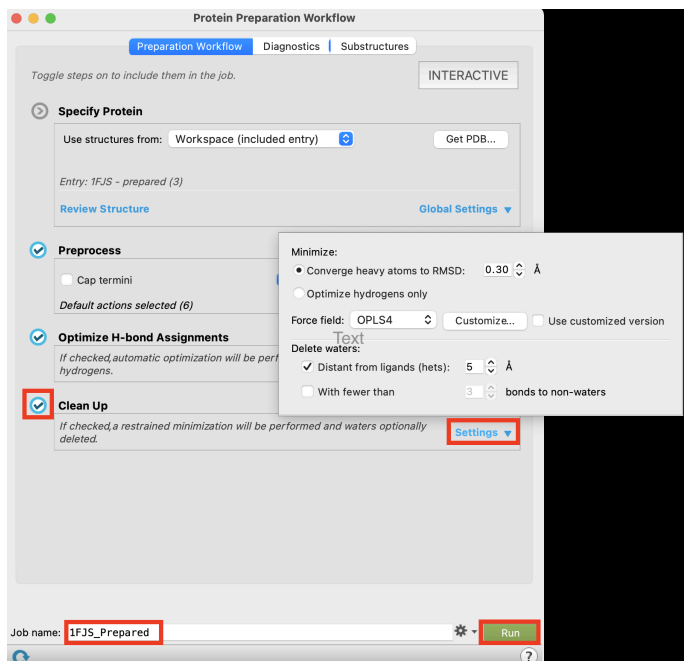
- Overlapping atoms caused by the addition of hydrogens during the **Preprocess** step (prev. step) will be corrected, and side chains may be flipped when this job is run.

Check the pH for **Optimization** ("Optimization uses:")

- This value should be captured in the pH range chosen during the **Preprocess** step (prev. step)

This step is crucial because even small changes in hydrogen positions can dramatically affect docking results. The optimized hydrogen bonding network provides a more realistic environment for ligand binding. When interpreting the results, pay attention to hydrogen bonds around the binding site—these will influence

ligand recognition and binding affinity predictions.



Clean Up default Settings

Confirm **Clean Up** is toggled on
In newer versions of Maestro the same functionality can be found by “**Minimize and Delete Waters**”.

This step involves energy minimizing the structure to relieve unphysical steric clashes introduced during hydrogen addition (by small atomic movements while maintaining the overall protein fold) + remove water molecules not involved in important interactions (non-essential).

Click **Settings**

- Notice that by default distant waters will be removed. The specified distance to ligands can be modified

Change the “Job name” below to **1FJS_Prepared**

Click **Run**

- This job will take a few minutes as all toggled jobs are running consecutively
- After this job will be done you'll see that banner appears with a message that **1FJS_Prepared** has been incorporated.
- New group will be added to the Entry List

As result you'll get relaxed protein structure with:

- Reduced steric clashes
- Lower energy conformation
- Only structurally or functionally relevant water molecules (5 ang. from protein, as you selected above)

The minimized structure has a more reasonable geometry than the initial structure. The RMSD value of 0.3Å indicates the constraint strength—allowing enough movement to relieve clashes but not enough to significantly alter the experimentally determined structure (protein folding). The removal of distant waters simplifies the system without sacrificing important water-mediated interactions near the binding site.

ENTRY LIST		
Row	In	Title
1	☐	1FJS
2	☐	1FJS - with-deletions
3	☒	1FJS - prepared

Select the protein preparation output

In the Entry List, click **1FJS - prepared**

Note: In **Interactive mode**, multiple entries will be added to the Entry List as each individual preparation task will be run as an independent job.

Protein Preparation Workflow

Preparation Workflow | Diagnostics | **Substructures**

Reload from Workspace Entry: 1FJS - prepared (3)

Choose items below to view in Workspace, copy, or delete. Select

Ligands, Metals, Other. The Lig column shows detected ligands. To change the classification, visit the [Ligand Detection...](#) settings, then click [Reload from Workspace](#) above.

The *Preprocess* step may generate multiple states for your ligands. The (likely) most favorable state will be checked by default. Optionally choose a different state to keep.

Lig	Chain	Res Name + #	S1	S2	S3	S4
	A	CL 508	☒			
X	A	Z34 500	☒	☐	☐	☐
	A	CA 507	☒			
	A	GOL 502				
	A	GOL 503				

Waters:

Chain	Res Name + #
A	HOH 602
A	HOH 604
A	HOH 605
A	HOH 606
A	HOH 607

Chains:

Chain	Type
A	Protein

42 items selected Clear Copy to New Entry **Delete from Entry**

Prepare Selected Only... < Diagnostics Workflow >

Return to the **Protein Preparation** tool

Click the **Substructures** tab
Choose **Reload from Workspace/Load workspace entry**

In the **Hets table** (“Ligands, Metals, Other”), shift-click to select all “GOL” rows

Click **Delete from Entry**

In the **Waters table**, shift-click to select all waters

Click **Delete from Entry**

Note: These waters could have been removed during the **Clean Up** processing step.

Depending on your system and research question, you may want to keep certain waters. See Protein Structure Preparation using the [Protein Preparation Workflow](#) or [Protein Preparation Workflow Panel Help](#) for more details.

Perform Substructures Review

While here we are just removing all waters, some water molecules in binding sites are conserved and crucial for ligand binding (e.g. bridging interactions in the binding site). Situation differs from one to another project and a decision about it you need to make taking into account specific project goals, protein of interest and ligands.

Protein Preparation Workflow

Preparation Workflow | **Diagnostics** | Substructures

Check Workspace Entry Entry: 1FJS - prepared (3)

No issues were found. See Reports for more information about the protein.

Valences Missing Overlapping Alternates **Reports**

View: Steric Clashes

Select table rows to review the corresponding items in the Workspace:

Atoms	Distance	Min Allowed	Delta
-------	----------	-------------	-------

Confirm issues have been resolved during preparation

In the Entry List in **1FJS_Prepared-out1** group, select **1FJS – with-deletions**

Return to the **Protein Preparation Workflow** window and click the **Diagnostics** tab to make sure there are no issues missed during the preparation (table should be empty).

You may need to click **Check Workspace Entry**.

Exit the Protein Preparation Workflow.

Note: If issues persist after preparation, perform specific interactive protein preparations on the modified protein with adjusted settings. The job type will depend on which problems were found.

Rename **1FJS – with-deletions** (inside of **1FJS_Prepared-out1** group) to **1fjs_prep_complex** so it can be more obvious for you for future usage. To do that use **Project Table**.

The absence of issues in the final diagnostic check confirms successful preparation. The structure now has proper chemistry, optimized interactions, and reasonable geometry—essential qualities for reliable docking results.

Advice:

The Entry List can become cluttered quickly. Use these strategies:

- Create groups for different stages of your workflow (Project Table: right-click → Group...)

- Name groups clearly (e.g., "Raw Structures," "Prepared Proteins," "Docked Ligands")
- Use the "Include/Exclude All" option (right-click on group) to toggle visibility

Ligand Preparation

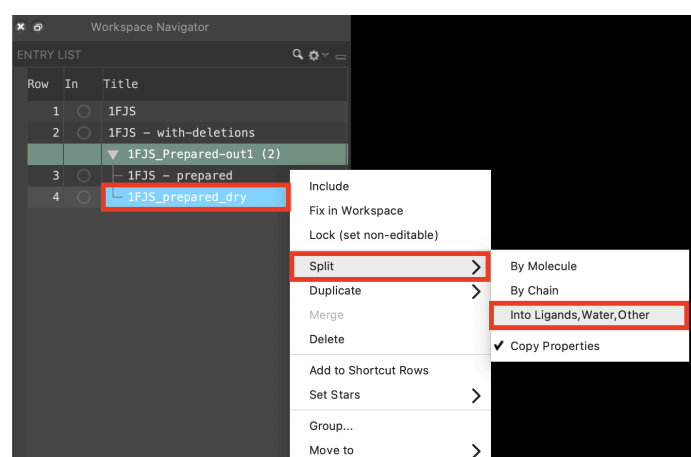
In this section, you will prepare the co-crystallized ligand from the 1FJS structure (= cognate ligand) for use in virtual screening. This is a typical step for cognate ligand docking, as it provides important validation prior to screening a larger ligand data set.

The following steps provide an example of how you would prepare a ligand data set using **LigPrep**. Ligand files can be sourced from numerous databases, often in the form of 1D or 2D structures with unstandardized chemistry. Before being used in a virtual screen, ligands must be converted to 3D structures, with their chemistry properly standardized and extrapolated.

During this step we're going to:

- Split protein-ligand complexes
- Run LigPrep on the cognate ligand
- Understand the process of multiple states generation (tautomers, protonation states)
- Prepare virtual screening ligands

Split the prepared structure



Right-click to split an entry into different components.

In the Entry List, double click **1fjs_prep_complex** in the **1FJS_Prepared-out1** group and rename the entry to **1FJS_prepared_dry**

Right-click on 1FJS_prepared _dry

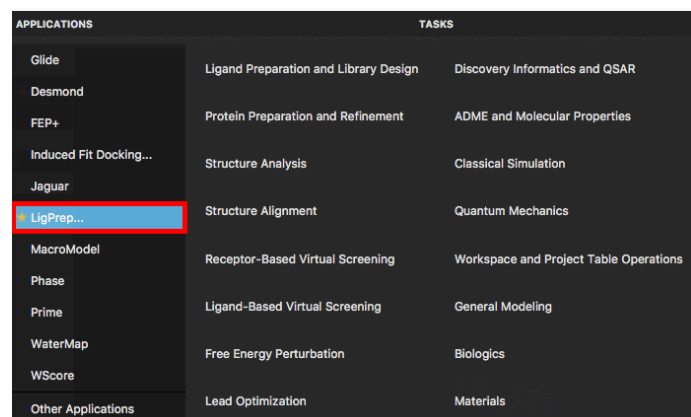
Choose **Split > Into Ligands, Water, Other**

- Two new entries appear in the Entry List:
1FJS_prepared_dry_ligand and **1FJS_prepared_dry_protein**.

Click on **1FJS – prepared_dry_ligand** and this way include only **1FJS – prepared_dry_ligand** in the Workspace (the entry is represented in the Workspace, the circle in the “In” column is blue)

- **Only** the ligand is displayed in the Workspace (without protein around it)

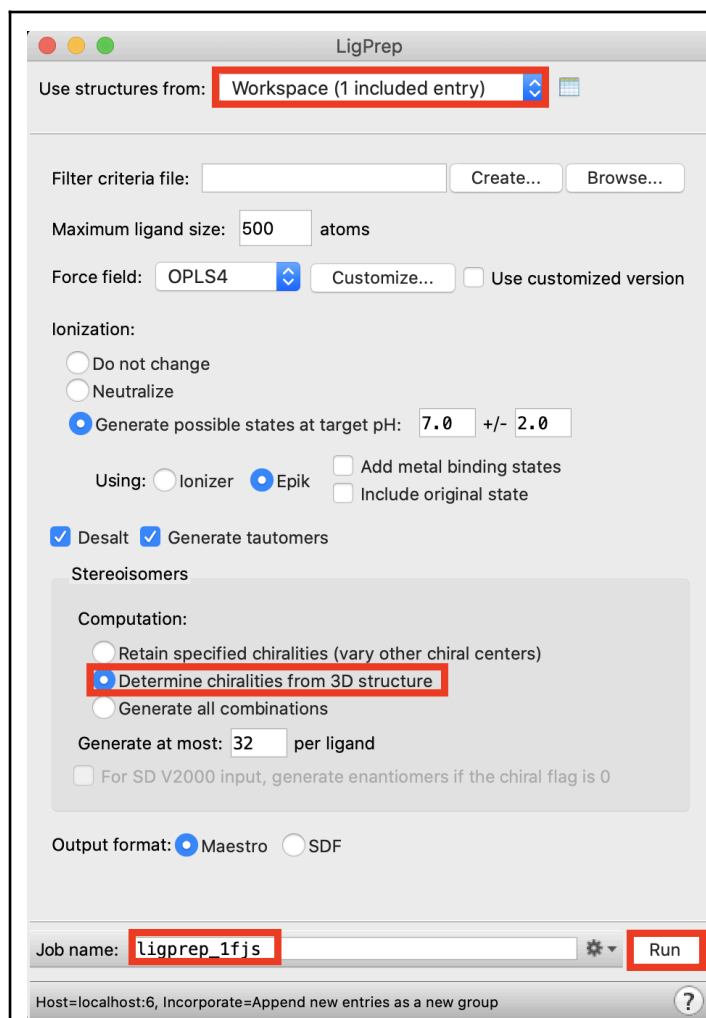
Run LigPrep



LigPrep application in the Task toolbar.

Go to **Tasks > Browse > LigPrep**

- The LigPrep panel opens



The LigPrep panel.

For **Use structures from**, choose **Workspace (1 included entry)**

In the “Generate possible states at target pH” section near “Using: ...” if there are 2 “Epik” options (“Epik Classic” and “Epik”) and one of them is already selected (whatever) - just leave it as it is.

Under **Stereoisomers**, choose **Determine chiralities from 3D structure**

Change **Job name** to **ligprep_1fjs**

Click **Run**

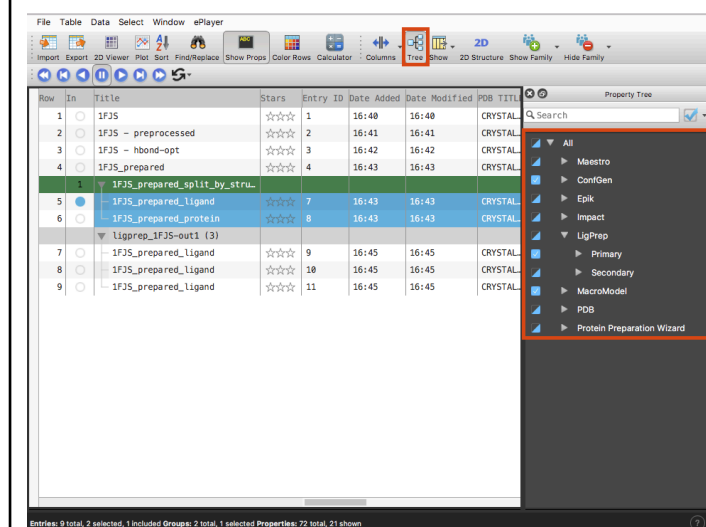
- A banner appears when the job has been incorporated
 - incorporated = once a job is finished, output files from the Working directory is added to the project and shown in the Entry List and Project Table
- A new group is added to the Entry List
- The number of ligands in this group is shown in parentheses

Note: The **Tile** functionality is very useful for seeing the slight variations in chemistry for the generated structures.

The **Tile View** can be turned on by clicking the **+** in the **Workspace Configuration Toolbar** in the bottom right corner and then clicking the **Tile** button.

You can open the **Workspace Configuration Toolbar** by clicking **Workspace > Configure...**

In the lower right corner appears a window, where you can click the **Tile** button.



The Project Table with the Property Tree open.

Type **Ctrl+T (Cmd+T)** to open the Project Table.

Click **Tree** button to open the **Property Tree**

- Different calculated properties can be toggled on and off
- Click the **arrow** next to each application to view more properties

Preparing a ligand using LigPrep may produce multiple output structures for each input structure. This changing number of ligands is completely normal in molecular modeling workflows. When Maestro's LigPrep processes your input ligands, it generates multiple versions of each molecule to account for their potential forms in biological environments:

- Multiple tautomers: For example, a molecule with a keto group (C=O) might exist as an enol form (C-OH) with the double bond shifted. Both versions get generated.
- Different protonation states: At physiological pH, some functional groups like carboxylic acids (-COOH) might exist as both protonated and deprotonated (-COO⁻) forms.
- Stereoisomers: If your input molecules have undefined stereochemistry at some positions, Maestro generates all possible configurations. For example, one carbon with undefined stereochemistry would produce two different structures.
- Ring conformations: Flexible rings (like cyclohexane) can adopt different 3D shapes (chair, boat, etc.) that get enumerated.

For example, a single molecule with one undefined stereocenter and the potential for two protonation states would generate 4 distinct structures (2 stereoisomers × 2 protonation states).

Question: Why is it important to prepare a ligand before proceeding with docking?

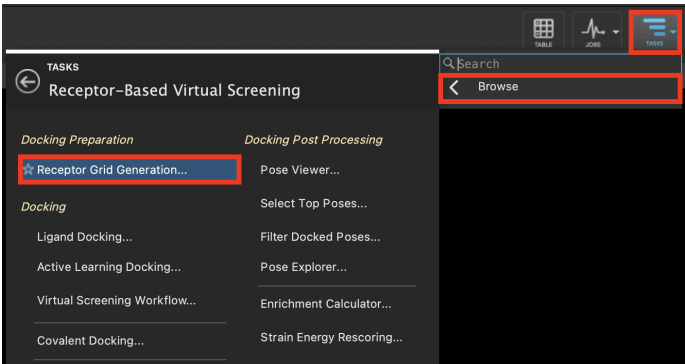
Task: Same way prepare all your ligands of interest. Don't forget to import your ligands of interest beforehand: **File > Import structures**. Afterwards you need to prepare them (same way as for cognate ligand).

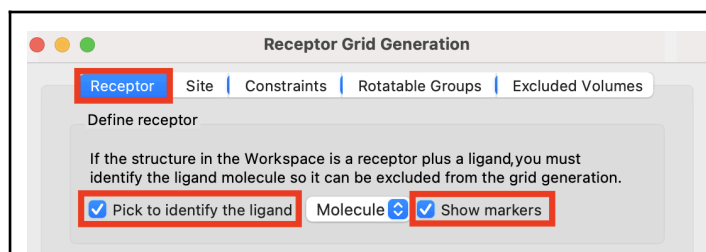
Receptor Grid generation

Grid generation must be performed prior to running a virtual screen with Glide. The grid defines the search space and properties for docking. The shape and properties of the receptor are represented in a grid by fields that become progressively more discriminating during the docking process. To add more information to a receptor grid, different kinds of constraints can be applied during the grid generation stage. For example, during next steps we will set a hydrogen bond constraint in our receptor grid.

During this step we're going to:

- Identify the binding site using cognate ligand
- Define the bounding box dimensions
- Set hydrogen bonding constraints (e.g., ASP 189)
- Run grid generation

Identify the binding site	
	Click the In circle next to 1fjs_prep_complex to include it in the <u>Workspace</u>  Click Presets > Apply Default Custom Preset - 1fjs_prep_complex is rendered using the Custom Preset Go to Tasks > Browse > Receptor-Based Virtual Screening > Receptor Grid Generation - The Receptor Grid Generation panel opens
<i>"Receptor Grid Generation" option in "Receptor-Based Virtual Screening".</i>	



The "Receptor" tab of "Receptor Grid Generation".

Under "Define Receptor", check the boxes for "Pick to Identify the ligand" (Molecule) and "Show Markers"

- A banner in the Workspace will prompt you to click on an atom in the ligand



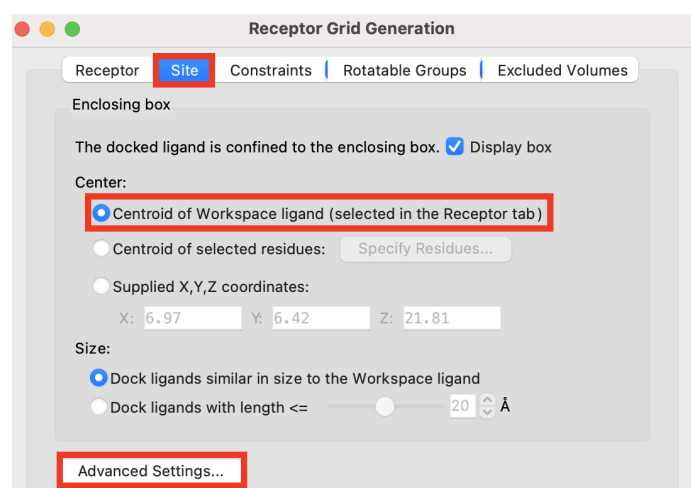
The ligand is defined to be excluded from grid generation.

Click on the **ligand** (presented as grey sticks with colorful atoms - quite noticeable in the Working space)

- The ligand is now highlighted with a purple box around it
- The ligand will be excluded from the grid generation

Note: The purple bounding box defines the region that the docked molecule(s) can occupy to satisfy the initial stages of docking

Define the bounding box dimensions



The Site tab of Receptor Grid Generation

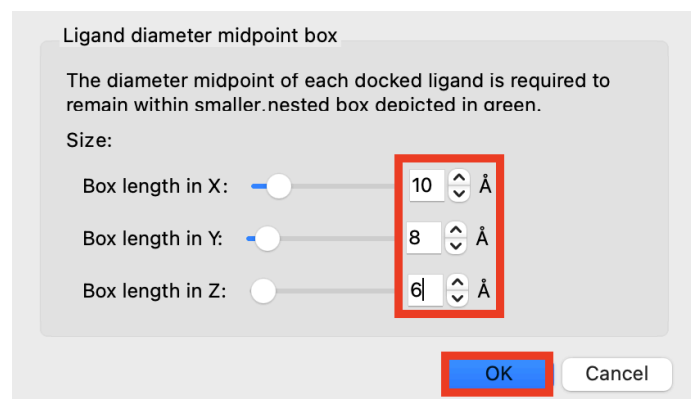
Click the **Site** tab

Select "Centroid of Workspace ligand (selected in the Receptor tab)"

Click "Advanced Settings..." button

- A green inner bounding box appears

Note: Appeared green bounding box (inside of the purple one) defines the region in which the centroid of the docked molecule(s) must occupy to pass the initial stages docking

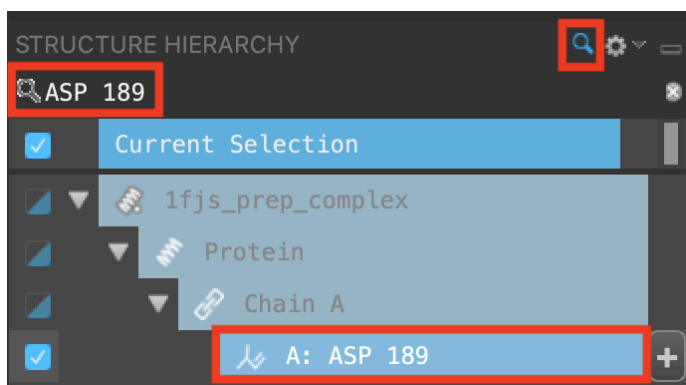


Adjust the settings for **X**, **Y**, and **Z** sizes to **10**, **8**, and **6 Å**, respectively.

- The shape of the green box is changed

Click **OK**

Set a hydrogen bonding constraint



Search in the Structure Hierarchy

Type **L** (on the keyboard) to zoom to the ligand

In the **Structure Hierarchy**, click the **magnifying glass**

In the search field, type **ASP 189**

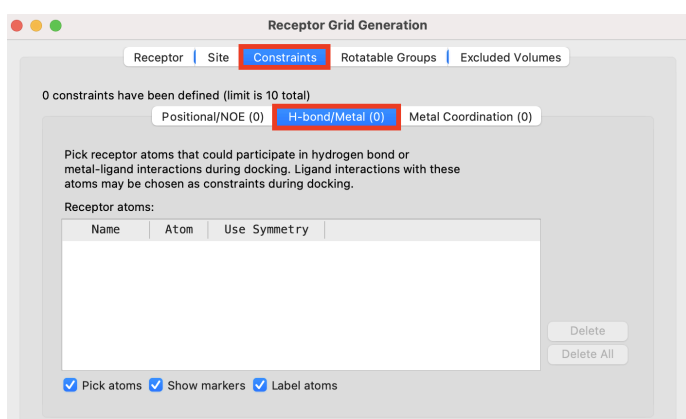
Select **ASP 189**

Note: Please see the [Introduction to Structure Preparation and Visualization](#) tutorial for instructions on how to add residue labels and show H-bonds



Zoom to selected atoms.

Under **Fit**, click **Fit view to selected atoms**

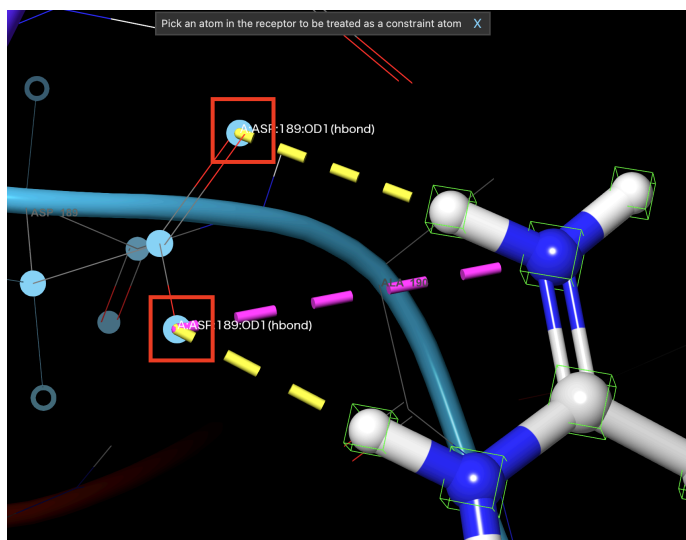


The Constraints tab of Receptor Grid Generation

In the **Receptor Grid Generation** panel, click the **Constraints** tab

Click the **H-bond/Metal (0)** tab

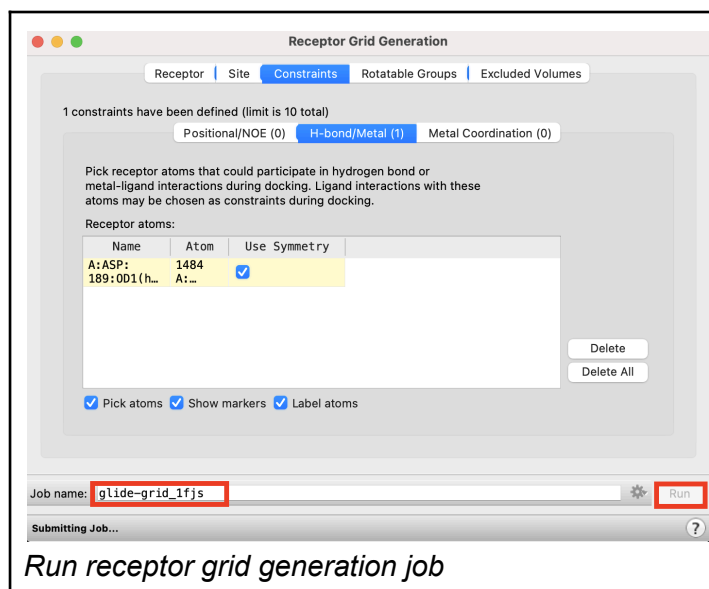
- A banner appears prompting selection of the receptor atom to be the constraint



Constraint defined on ASP 189

Click an **oxygen atom of the ASP 189 sidechain**

- Both oxygens are highlighted
- An H-bond constraint is defined in the Receptor atoms table



Change **Job name** to **glide-grid_1fjs**

Click **Run**

- This job will take about a minute
- At the end of the job banner appears.
- A folder named **glide-grid_1fjs** is written to your Working Directory

Now close the “Receptor Grid Generation” window and go to the next step.

Question:

Why is it necessary to generate a receptor grid? What would happen if you proceeded with docking a ligand without a receptor grid?

Docking the Cognate ligand (validation) and Screening Compounds (analysis)

The minimum requirements for running a Glide virtual screen are a **grid file** and a **ligand file**.

It is strongly recommended that the grid file be generated from a protein prepared using the [Protein Preparation Workflow](#) and the ligand file be prepared using [LigPrep](#). Additionally, you can choose the scoring function, set ligand- and receptor-based constraints, and define the output.

First, we will dock the cognate ligand, which is a helpful way to benchmark a virtual screen of compounds with unknown binding activity against a target. This crucial validation step ensures your workflow is correctly configured before screening many compounds. Only after validation should you dock your screening compounds.

Second, we will dock the screening compounds from a prepared ligand file, **50ligs_epik.mae.gz** (don't forget to upload these structures). Both jobs will use the receptor grid file that was generated in the previous step.

During **validation** step we're going to:

- Set up cognate ligand docking
- Apply constraints
- Run the docking calculation
- Analyze how well the docked pose matches the crystal structure

During **analysis** step we're going to:

- Set up screening compounds for docking
- Run docking calculations
- Understand why ligand numbers may decrease after docking

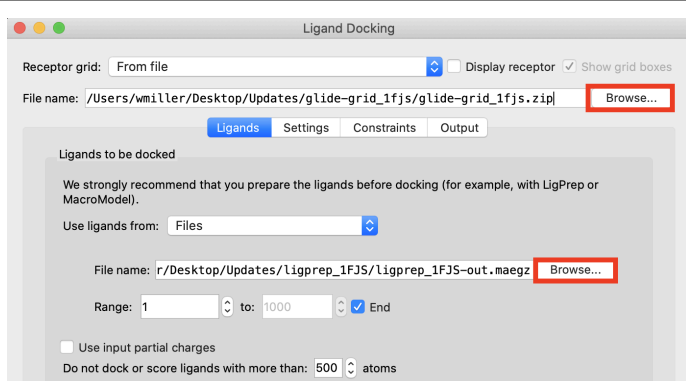
ENTRY LIST

Row	In	Title
1	☒	1fjs_prep_lig
2	☐	1fjs_prep_receptor
3	☐	1fjs_prep_complex

Include **1fjs_prep_lig** (in our case it's **1FJS_prepared_dry_ligand** (inside of **1FJS_prepared_dry** group and **1FJS_prepared_dry_ligand_split_by_structure** group)) in the Workspace (the entry is represented in the Workspace, the circle in the "In" column is blue)

Dock the cognate ligand

The information gained from this step can help with evaluating poses and beneficial interactions, which is useful for hit finding.



The Ligands tab of the Ligand Docking panel.

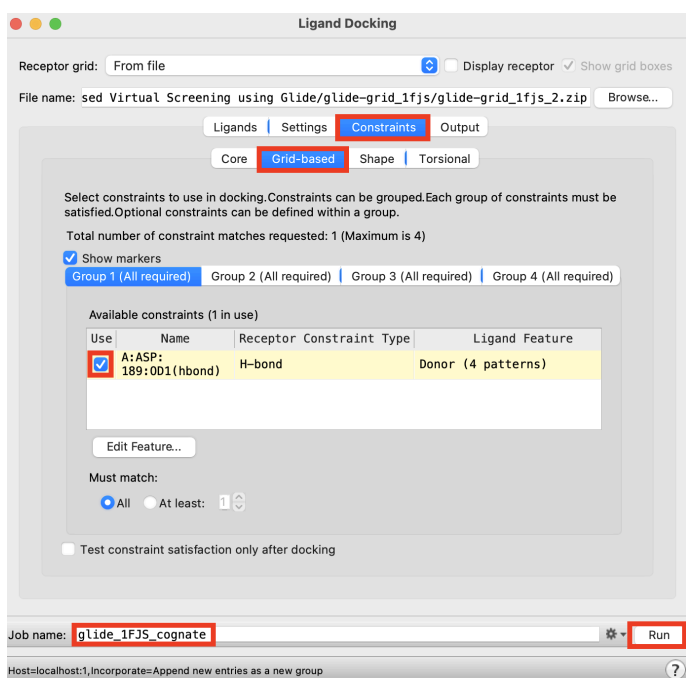
Go to **Tasks > Browse > Receptor-Based Virtual Screening > Ligand Docking**

- The Ligand Docking panel opens

Next to Receptor grid, click **Browse** and choose **glide-grid_1fjs.zip**

In the **Ligands** tab, for "Use ligands from", choose **Files**

Next to "File name", click **Browse** and choose **ligprep_1FJS-out.maegz**



The Constraints tab of the Ligand Docking panel.

Click the **Constraints** tab

Click on the **Grid-based / Receptor** tab

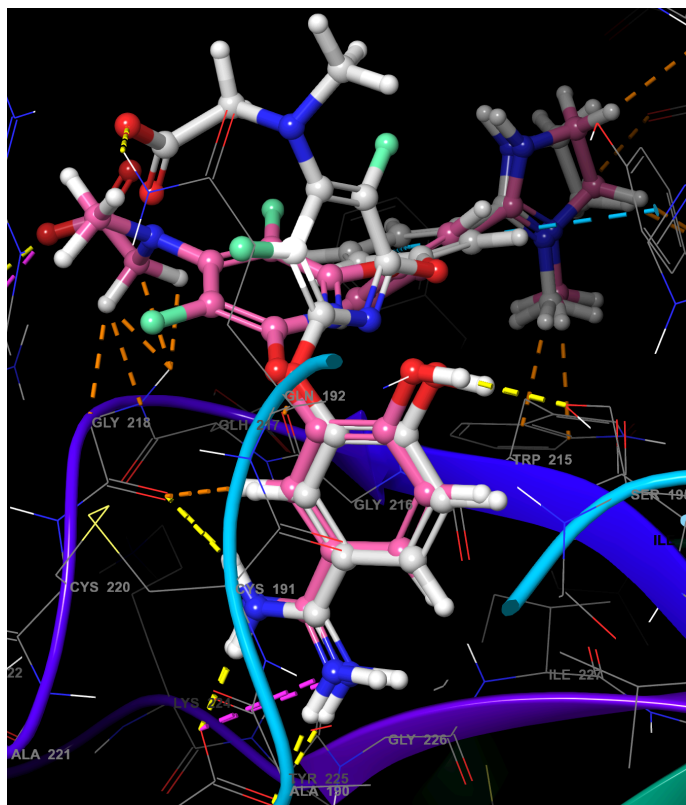
Under "Use" column in table, check the H-bond constraint for **ASP 189**

Change **Job name** to **glide_1FJS_cognate**

Click **Run**

- This job takes about a minute
- A banner appears to show that files have been incorporated
- A new group is added to the Entry List

Note: The **1fjs_prep_complex** entry is fixed in the Workspace, the top 1fjs_prep_lig entry is included, and the **Pose Viewer** panel appears.



Include other **ligand results**

- H-bonds to ASP 189 are conserved

Double-click **Presets**

Double-click the “In” circle next to

1fjs_prep_complex

- The entry is no longer fixed in the Workspace

Note: Though only the top ranked result is in strong agreement with the crystallographic pose, all three results accurately capture the pose of the ligand in the binding site (with varying degrees of success in capturing the solvent exposed region)

Binding pose of the top docked cognate ligand (pink) compared to the crystal structure (gray).

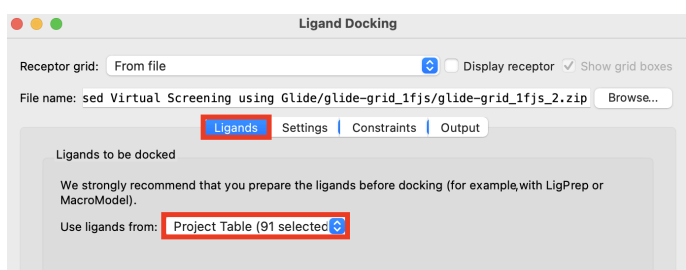
Dock the screening compounds

After importing and preparation steps all your ligands of interest will be within a separate group (let's name it **50ligs_epik**).

ENTRY LIST		
Row	In	Title
1	☐	1fjs_prep_lig
2	☐	1fjs_prep_recep
3	☒	1fjs_prep_complex
	☒	50ligs_epik (91)
4	☐	15650 Known Active
5	☐	15650 Known Active

Select **50ligs_epik** in the Entry List.

In the Entry List, select the group **50ligs_epik**

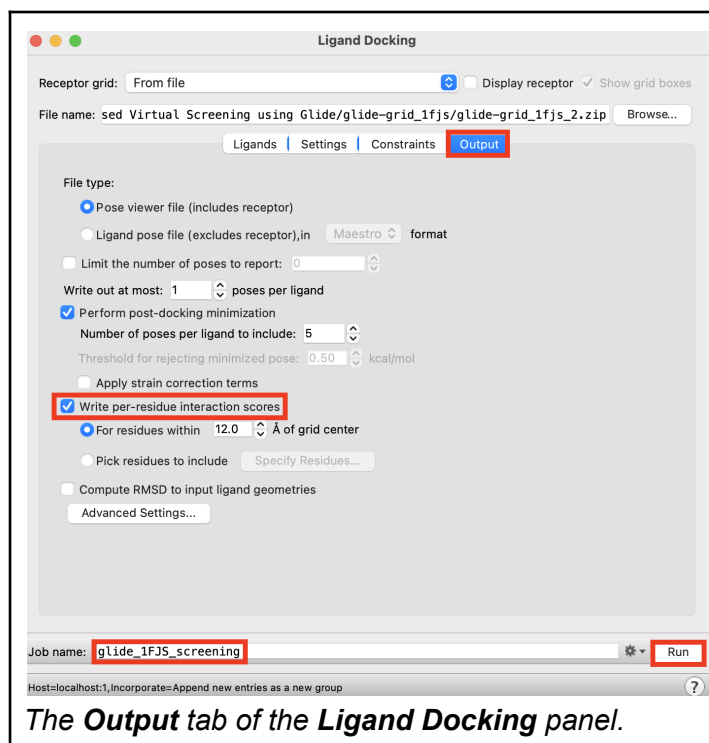


Use ligands from selected entries.

In the **Ligand Docking** panel, click the **Ligands** tab

For “Use ligands from”, choose **Project Table (selected entries)**

Note: Keep **glide-grid_1fjs.zip** as the receptor grid (Receptor grid, File name: ...)



Click the **Output** tab

Check **Write per-residue interaction scores**

Change “Job name” to **glide_1FJS_screening**

Click **Run**

- This job takes a few minutes
- A banner appears to show that files have been incorporated
- A new group is added to the Entry List

After docking, the number of ligands can decrease for several reasons:

- Docking failures: Some ligands may be too large for the binding site or have steric clashes that prevent successful docking.
- Score-based filtering: Maestro typically keeps only the best-scoring pose for each ligand, and may further filter out low-scoring compounds altogether.
- Assigned constraints (in our case: H-bond with ASP 189).
- Post-processing filters: The workflow might automatically apply criteria like:
 - Requiring specific interactions with key protein residues
 - Rejecting poses with unfavorable energy profiles
 - Filtering based on pose clustering
- Redundancy elimination: Some conformations might be so similar that the program consolidates them.

The final amount of ligands represent the most promising subset of your original library based on their predicted binding to your target protein.

Basic results analysis and binding-site characterization

Multiple Glide docking results can be viewed in the Entry List and be identified by the job name. Docked results will show the receptor in the first row (**1fjs_prep_complex**) and the docked ligand(s) in the subsequent row(s), where they are ordered by best to worst docking score, or Glide Gscore if Epik state penalties were not applied in LigPrep.

The Glide Gscore is broken down by van der Waals electrostatic components and can be seen in the Project Table, using the Property Tree. You can read more about how docking scores/poses are generated [here](#) and [here](#) and what dependencies they have [here](#) and [here](#).

The Entry List will show your receptor (protein) and docked ligands ordered by docking score. The Glide Gscore represents binding affinity (more negative = stronger, better binding). The score is broken down into components (van der Waals, electrostatic interactions, etc.). The breakdown helps you understand which interaction types contribute most to binding. This is crucial for drug design as you can optimize molecules based on these interactions.

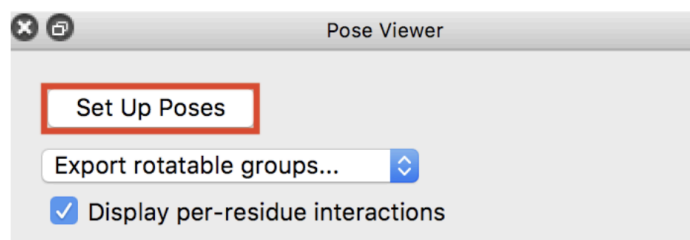
During this step we're going to:

- Use **Pose Viewer** to visualize docked poses

- Analyze the Project Table data
- Understand scoring metrics (docking score, Glide gscore, emodel)
- Export data for presentation

Visualize the results using Pose Viewer

Pose Viewer is a tool to visually examine how ligands are positioned in the binding site. Visual inspection is essential to understand the spatial arrangement of ligands and their interactions with specific protein residues.



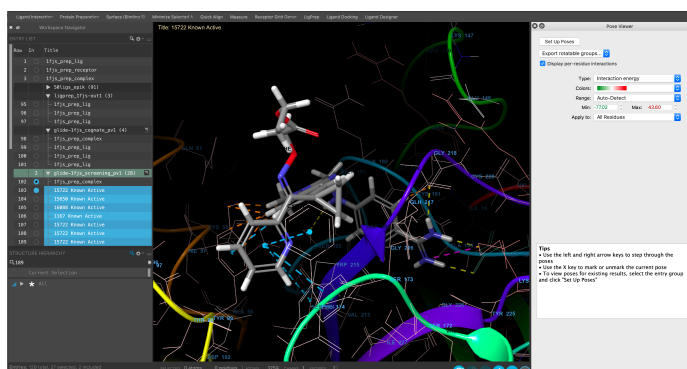
Pose Viewer panel.

Go to **Tasks > Browse > Receptor-Based Virtual Screening > Pose Viewer**

Select the newly generated group titled ... (contains newly generated items for cognate ligand)

Click **Set Up Poses**

Check **Display per-residue interactions**



Pose Viewer panel.

Step through the results using the **right** and **left arrow keys**

- Ligand poses are displayed in the Workspace
- Color-coded protein residues showing interaction energies: residues are colored according to their interaction energies, ranging from **green** (negative, favorable) to **red** (positive, unfavorable)

Task: look for consistent interactions across multiple good-scoring ligands, which may indicate important binding features. Pay attention to green residues as they contribute favorably to binding. This visual data complements the numerical scores.

Close the **Pose Viewer** panel

Analyze the results in the Project Table

That's the way to organize and view numerical data about your docking results. You need it to compare multiple ligands objectively based on their scoring parameters and to prepare data for presentation and further analysis.

Row	In	Title	docking
1		ifjs_prep_lig	
2		ifjs_prep_receptor	
3		ifjs_prep_complex	
		50ligs_epik (92)	
		ligprep_IFJS-out1 (3)	
		glide_IFJS-cognate_pv1 (4)	
2		glide_IFJS_screening_pv1 (2...	
103		ifjs_prep_complex	
104		15658 Known Active	
105		15722 Known Active	
106		1167 Known Active	
107		16088 Known Active	
108		15722 Known Active	
109		15722 Known Active	

Glide Primary properties shown in the Project Table.

In the Project Table, click the **Property Tree** icon

- The **Property Tree** appears on the right of the Project Table

Click the **All** checkbox twice

- All boxes are deselected

Click the **Glide** box

Click **Primary**

- Only the **Glide Primary** properties are shown

If you want to store this data: **Data > Export > Spreadsheet.**

Task: analyse all the data that is provided in **Spreadsheet**. Find out what each column means.

	Hint: in this Spreadsheet you can find a breakdown that helps you understand which interaction types contribute most to binding. The table shows various scoring metrics that help you rank compounds. Remember: for your presentation on the independent work carried out, you'll need to explain what each column represents (docking score = overall binding energy estimate, Glide gscore = modified score that may include penalties, emodel = extended model score that emphasizes good geometric positioning). <u>Note:</u> Please see Knowledge Base Article 1027 for more information on the difference between docking score, Glide gscore, and glide emodel score.
--	---

Questions:

1. What important protein-ligand interactions do you see when the cognate ligand is docked? List specific residues and define specific interactions that may play an important role in binding.
2. Select your top 5 ligands with the best docking scores. Remember that the lower the value, the better the docking score.
For each ligand:
 - i) write the ligand name and take a screenshot of the pose,
 - ii) list the docking score that can be found in the Project Table
 - iii) identify at least 2 types of protein-ligand interactions between that particular ligand and residues within the active site.
3. SiteMap visualization uses a grid of points to identify potential hydrophobic and hydrophilic regions; the hydrophilic regions are further classified into hydrogen-bond donor, hydrogen-bond acceptor, and metal-binding regions, and the surface of the protein is contoured. Take a screenshot of your SiteMap results. Identify which regions of your receptor are hydrophilic and hydrophobic.

Advanced Binding Site analysis with SiteMap

We will analyze the binding site using **SiteMap**.

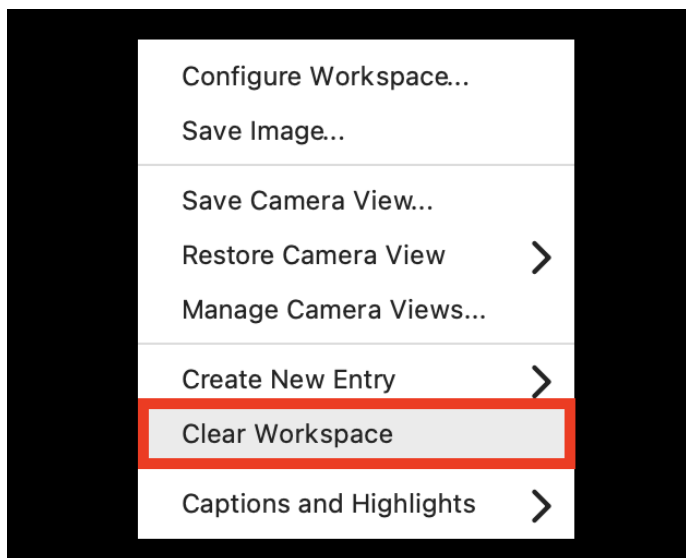
SiteMap characterizes hydrophilic, hydrophobic, acceptor, and donor regions of a receptor. This is useful for learning more about an active site, predicting a binding site in an apo structure, or identifying possible allosteric sites. SiteMap ranks the potential binding sites with a druggability score, which can be viewed in the Project Table. The output from a Glide virtual screen can be overlaid with SiteMap information to examine how well the docked ligands explore the various regions in the binding cavity. Sites identified by SiteMap can also be used to create receptor grids for virtual screening experiments. This can be useful for exploring sites without a known active compound.

During this step we're going to:

- Run SiteMap to characterize binding site properties
- Interpret hydrophobic / hydrophilic regions
- Create a receptor grid from SiteMap results
- Use SiteMap for finding potential allosteric sites

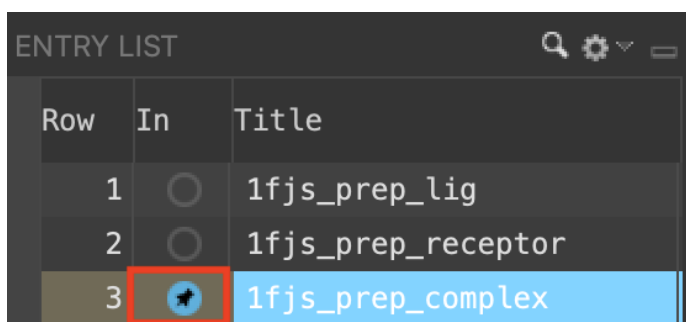
Identify a binding site with SiteMap

SiteMap is a tool that characterizes the properties of binding sites on a protein's surface. You need it to understand the physical and chemical properties of binding sites, which is crucial for evaluating their druggability and for designing complementary ligands.



Clear Workspace.

Right-click an empty area in the Workspace choose **Clear Workspace**

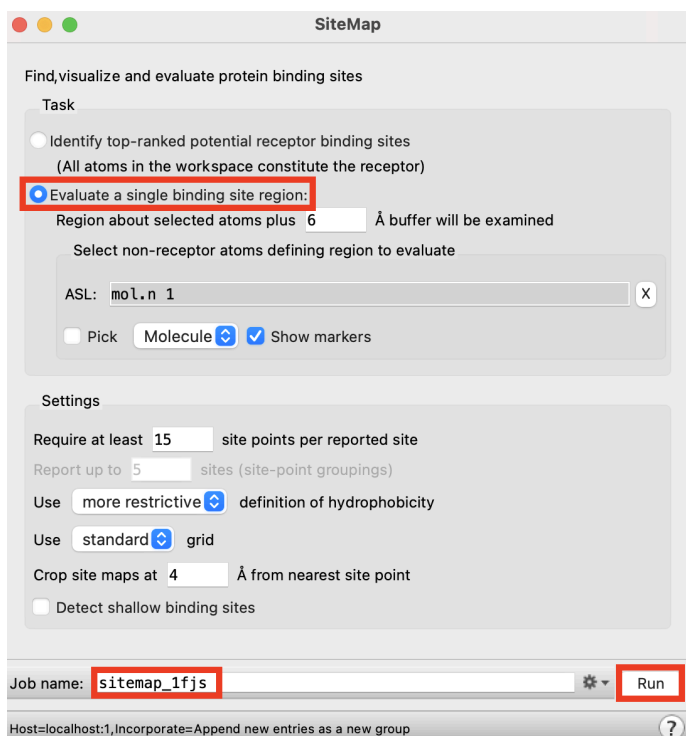


Fix (pin) 1fjs_prep_complex in the Workspace.

Double-click the **In** circle to fix **1fjs_prep_complex** (in row 3) in the Entry List

Go to **Tasks > Browse > Structure Analysis > Binding Site Detection**

- The SiteMap panel opens



SiteMap panel.

Under Task, select **Evaluate a Single binding site region**

As molecule select ASP 189 (as before)

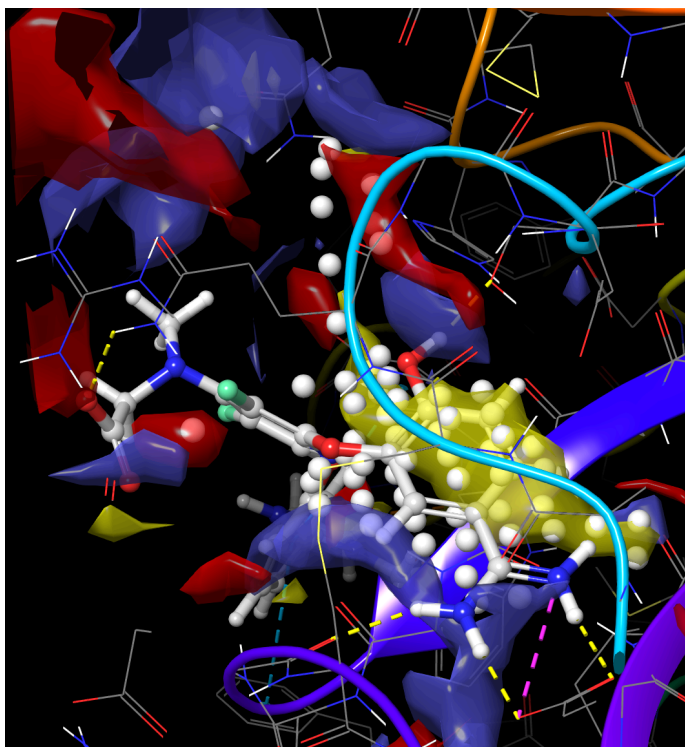
Click on the **ligand** in the Workspace

- The **ligand** is highlighted
- SiteMap removes the **ligand** from the calculation

Change the "Job name" to **sitemap_1fjs**

Click **Run**

- A banner appears when the job has incorporated (once a job is finished, output files from the *Working Directory* are added to the project and shown in the Entry List and Project Table)
- A new group is added to the Entry List



SiteMap results in the Workspace.

Include the **sitemap_1fjs_site_1**, **sitemap_1fjs_ligand**, and **1fjs_protein** (included items have filled the blue circle in the “In” column)

Type L

- Various surfaces are shown representing different regions of hydrophilic property; **hydrophobic** regions (yellow surfaces), **hydrogen bond acceptor** regions (red surfaces), **hydrogen bond donor** regions (blue surfaces)
- The **white** site-point spheres each represent ~1 Å³ volume elements that define the site

In the Entry List, click the **S** next to **sitemap_1fjs_site1** to **toggle** the surfaces associated with the SiteMap

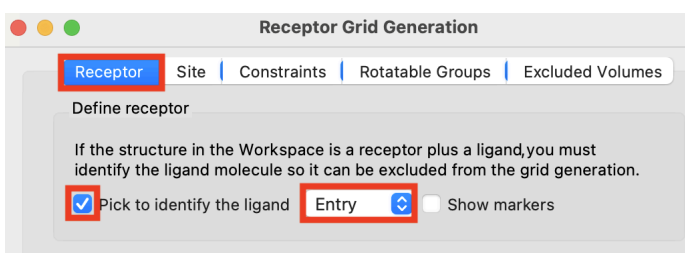
Note: To find all possible binding sites using SiteMap, under **Task** select **Identify top-ranked potential receptor binding sites**.

If you want to detect all pockets, you will need to exclude the cognate ligand from the Workspace.

The colored surfaces show what types of chemical groups would interact favorably with different regions of the binding site. For example, hydrophobic ligand groups should align with yellow surfaces, while hydrogen bond donors should align with red surfaces. A good binding site typically has a balance of these properties in a well-defined pocket.

Generate a Receptor Grid from SiteMap

That's creating a computational grid that defines the binding site for future docking calculations. You need it because his grid becomes the target for virtual screening of compound libraries. It translates the visual information from SiteMap into a format that docking algorithms can use.



The Receptor tab of Receptor Grid Generation.

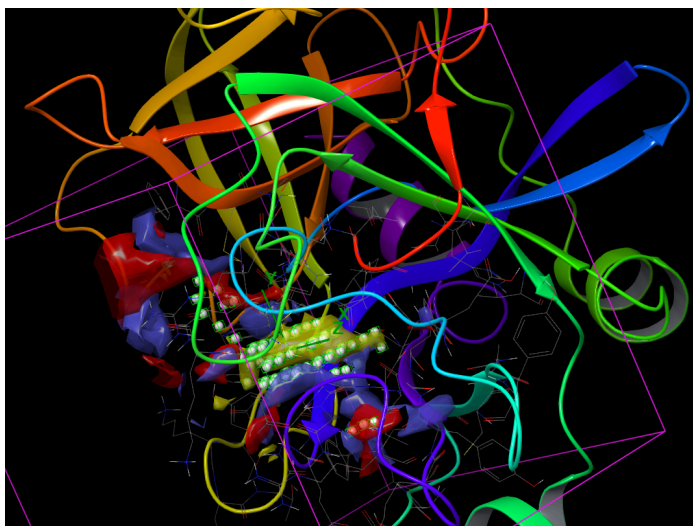
Include only the **sitemap_1fjs_site_1** and **sitemap_1fjs_protein** (included items have filled the blue circle in the “In” column)

Go to **Tasks > Browse > Receptor-Based Virtual Screening > Receptor Grid Generation**

Click the **Receptor** tab

Check **Pick to identify ligand** and choose **Entry**

- A banner appears prompting to pick an atom



Click on a white **site point** in the Workspace

- All site points are highlighted

A receptor grid using SiteMap to define a potential binding site.

Change “Job name” to **glide-grid_1fjs_sitemap**

Click **Run**

- This job takes about a minute
- A folder named **glide-grid_1fjs_sitemap** is written to your Working Directory

Result: computational grid defining the binding site. It can be used further for subsequent docking studies. The grid itself is a computational construct used by Glide for docking calculations. The importance is that it's centered on the binding site you identified with SiteMap, ensuring that future docking studies focus on the region of interest.

The Receptor tab of Receptor Grid Generation.

Advanced visualization techniques for protein-ligand complexes

In this section, you will explore ways to visualize structures in the Workspace. Object representation can be changed in a number of ways using the **Style toolbox**.

Presets offer the ability to quickly render a structure in a number of styles, similar to PyMOL, to facilitate easy visualization.

Presets can be used in a variety of ways, from de-cluttering your structure to creating publication-quality images.

You will analyze the protein-ligand complex by looking at the interactions, and generate a custom set for some binding residues of interest.

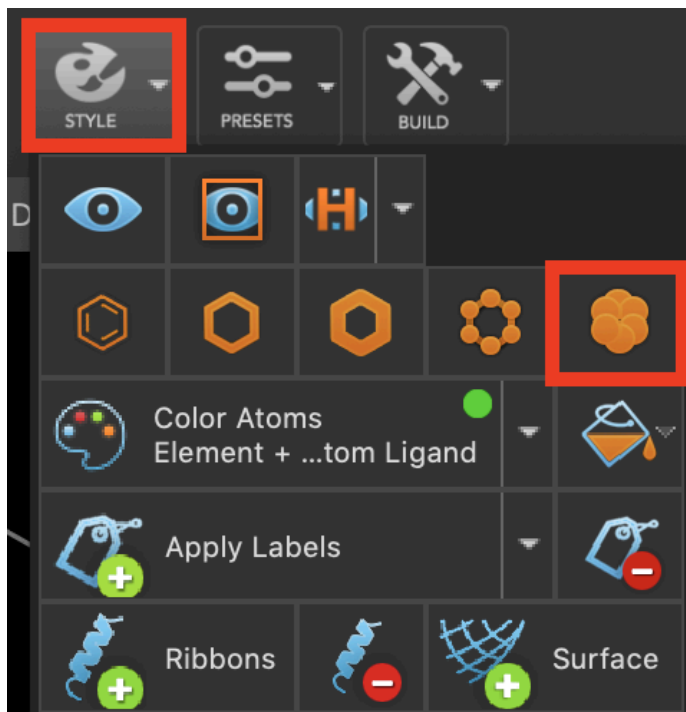
Finally, you will visualize the surface of the binding pocket and save an image of the complex.

During this step we're going to:

- Use the **Style toolbox** for protein-ligand visualization
- Apply presets for better visualization
- Visualize interactions

- Create custom sets for important residues
- Generate and manipulate surfaces
- Create 2D interaction diagrams
- Visualize XP results
- Save high-quality images

Use the Style toolbox



The Style toolbox with CPK representation highlighted.

Include entry **1FJS – prepared**

- the entry is represented in the Workspace, the circle in the “In” column is blue

Type **L**

- The Workspace zooms to the ligand

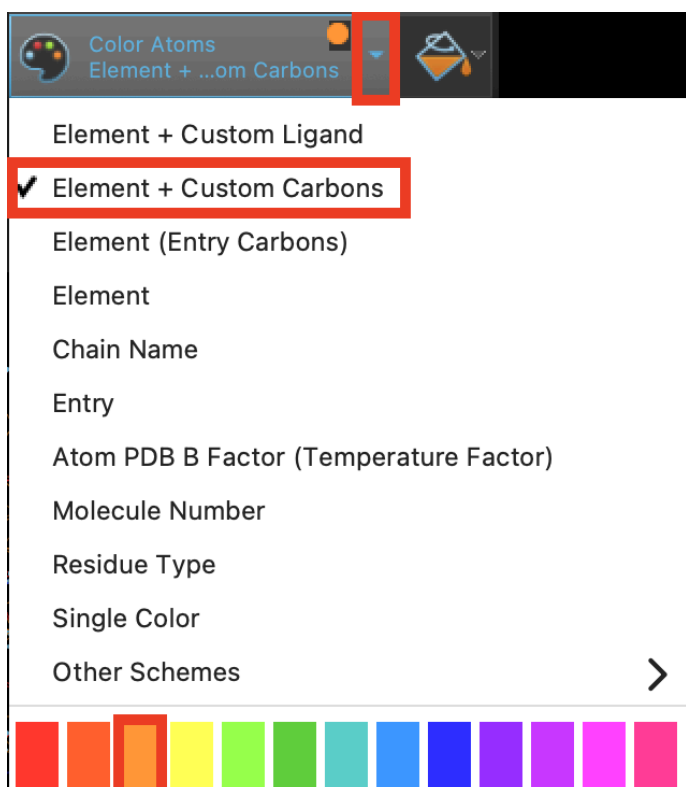
Under **Quick Select**, click **L**

- The ligand is selected:
 - (1) the atoms are chosen in the Workspace. These atoms are referred to as "the selection" or "the atom selection". Workspace operations are performed on the selected atoms.
 - (2) The entry is chosen in the Entry List (and Project Table) and the row for the entry is highlighted. Project operations are performed on all selected entries)

Click **Style**

Choose **CPK** representation

- The ligand is rendered in space-filling (CPK) representation
- This is only applied to the ligand, since nothing else is selected in the Workspace



The Color Atoms menu.

Click the “**Color Atoms**” arrow

Choose **Element (Custom Ligand)**, and pick orange from the secondary menu

- Ligand carbon atoms are orange

Under **Quick Select**, click **P**

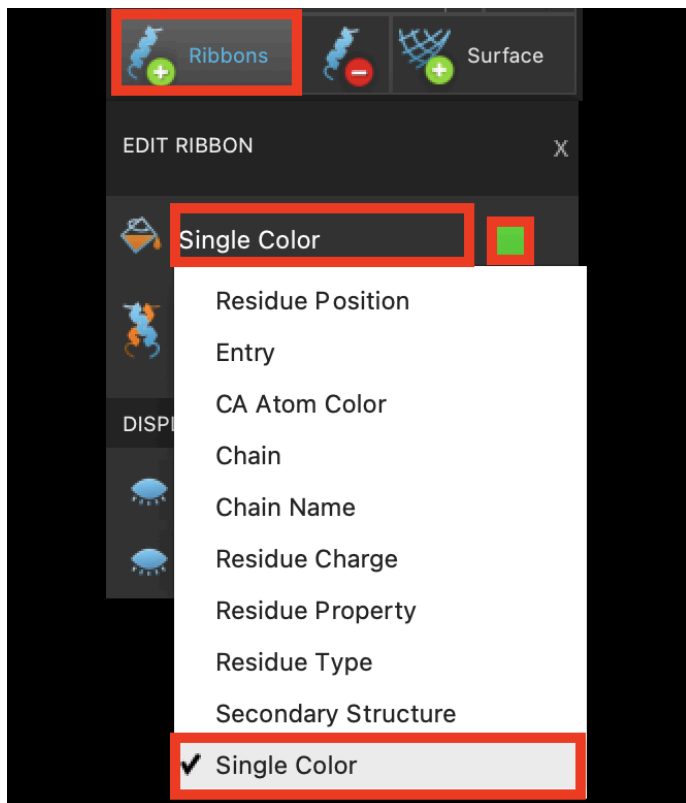
- The protein is selected
 - (1) the atoms are chosen in the Workspace. These atoms are referred to as "the selection" or "the atom selection". Workspace operations are performed on the selected atoms.
 - (2) The entry is chosen in the Entry List (and Project Table) and the row for the entry is highlighted. Project operations are performed on all selected entries)

Type **Z**

- The Workspace is zoomed to view the selected structure: protein that is selected

In the **Style toolbox**, click **Ribbons**

- Ribbons are added to the protein



The Edit Ribbon panel.

Right-click on the **Ribbons**

- The **Edit Ribbon** panel opens

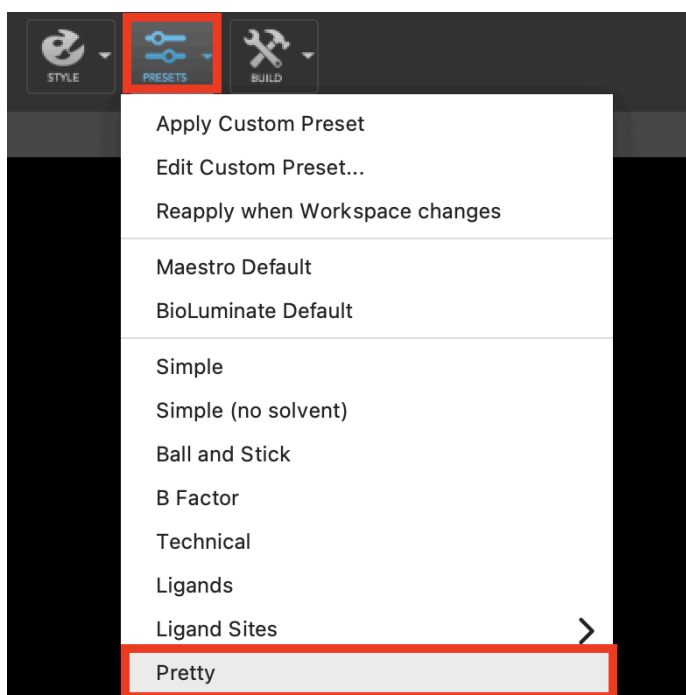
Note: Use the predictive highlighting to know when you will click on the ribbon.

Click **Residue position** in the color scheme

Choose **Single Color**

Note: Click the box to the right of the color scheme to choose different colors

Apply a Preset style



The Presets menu.

Click **Presets**

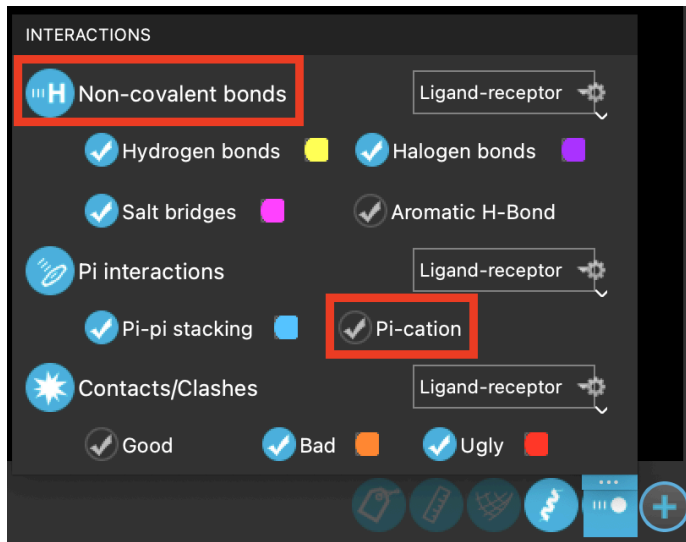
Choose **Pretty**:

- The Workspace is rendered with ribbons,
- a green thick-tube ligand,
- side chains are hidden.

Double-click **Presets**

- The Workspace is redrawn with the **Custom Preset**
- The Workspace zooms to the ligand

Visualize Interactions



The Interactions panel in the Workspace Configuration Toolbar.

In the **Workspace Configuration Toolbar**, right-click **Interactions**

- The Interactions panel opens

Turn on **Non-covalent bonds**

Turn off **Pi-cation interactions**

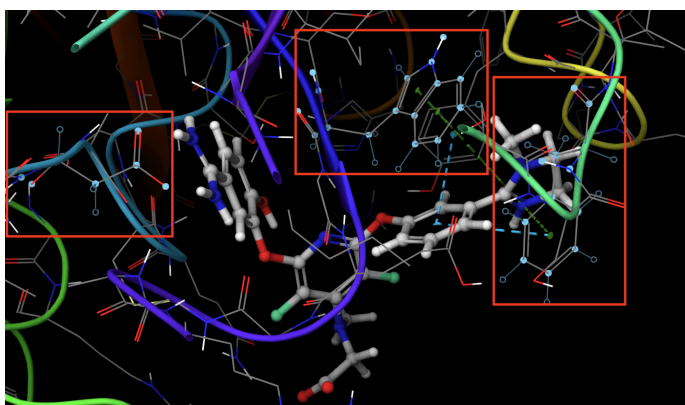
Note: Clicking the color to the right of each interaction opens the **Preferences** panel, where the interaction visualization can be customized

Note: The threshold for Contacts/Clashes is set to

- 0.89 for bad
- 0.75 for ugly.

These values correspond to the ratio of the distance between the two atoms and the sum of their Van der Waals radii.

Create a custom set

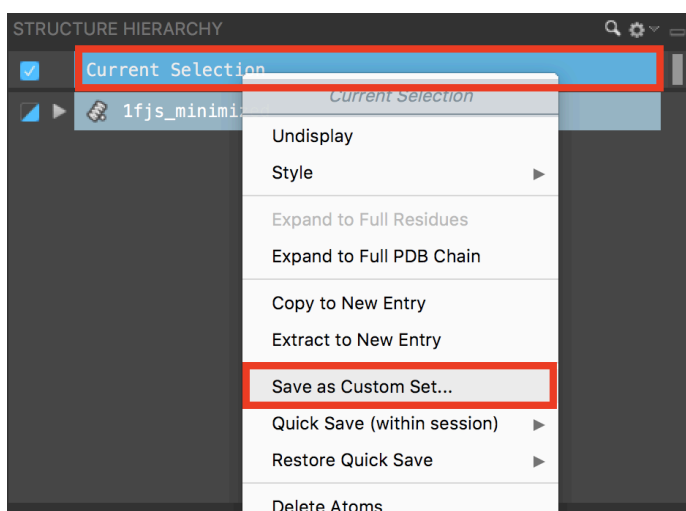


Select PHE 174, TRP 215 and ASP 189.

Type **R** to switch to residue picking

Ctrl+Click to select the binding site residues PHE 174, TRP 215 and ASP 189 in the Workspace

Note: Residues can be located by residue number and type in the **Structure Hierarchy**

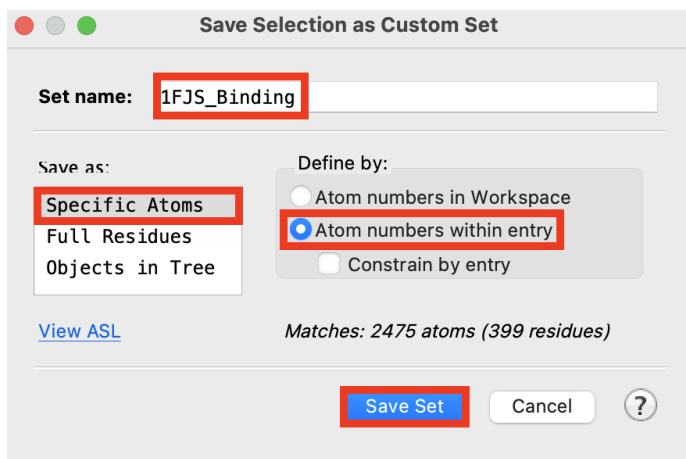


Open "Save as Custom Set" panel.

In the **Structure Hierarchy**, right-click on **Current Selection**

Choose **Save as Custom Set**

- The **Save Selection as Custom Set** panel opens



Save “Custom Set”.

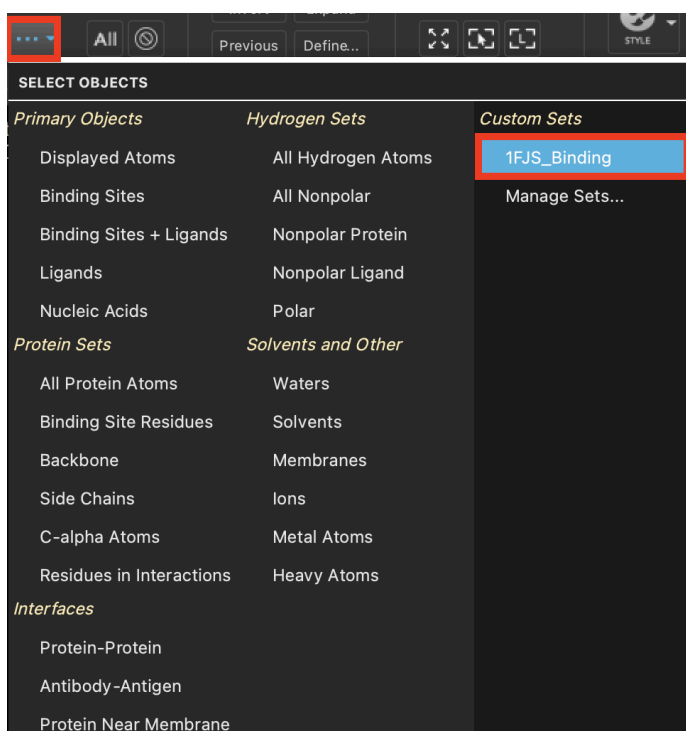
For **Set Name**, type **1FJS_Binding**

For **Save as**, choose **Specific Atoms**

For **Define by**, choose **Atom numbers within entry**

- This will ensure that the selection will remain consistent when multiple entries are in the Workspace

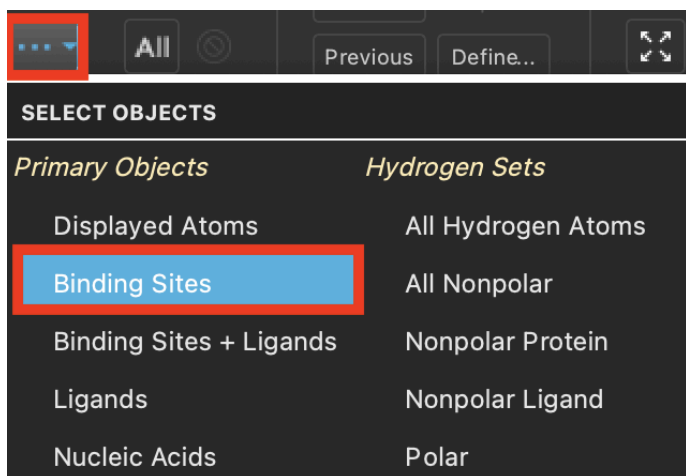
Click **Save Set**



“Custom Sets” section in the “Selection Toolbar” dropdown.

Note: Custom Sets can be accessed and edited through the **Custom Sets** section in the dropdown in the **Selection Toolbar**

Generate and manipulate a surface



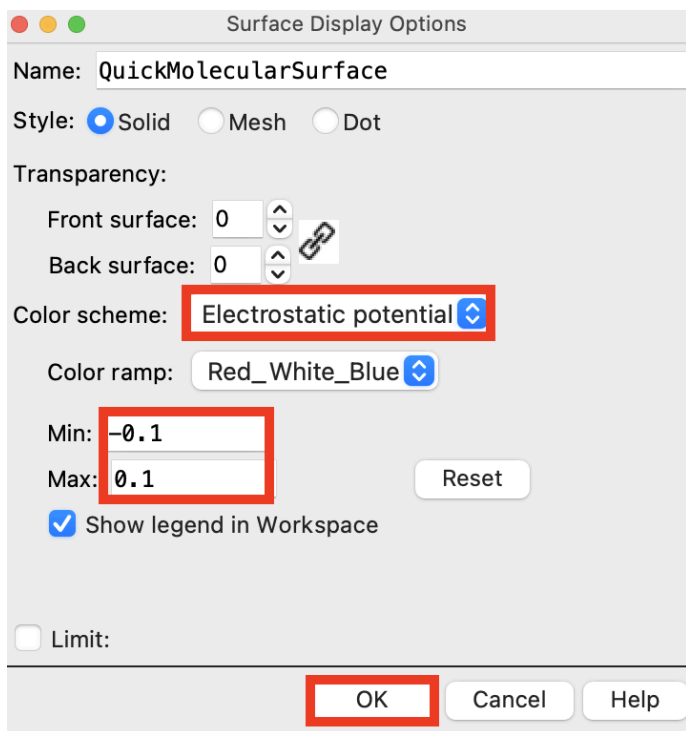
More options in Quick Select.

Under **Quick Select**, click “...” and choose **Binding Sites**

Click **Style** and choose **Surface**

- A solid gray surface is applied
- An S is next to the title in the Entry List, click to see surface options

Note: Click **Surface (Binding Site)** in the **Favorites** toolbar to perform the same task



Right-click the **Surface**

Choose **Display Options**

- The Surface Display Options panel opens

For **Color Scheme**, choose **Electrostatic Potential**

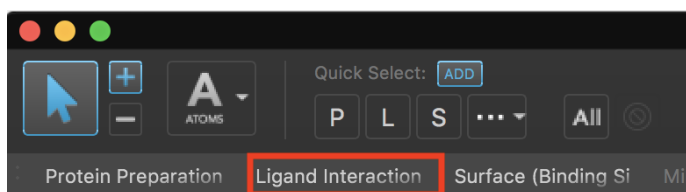
Change the **Min** and **Max** values to **-0.1** and **0.1**, respectively

Click **OK**

- The intensity of the surface colors is increased

The Surface Display Options panel.

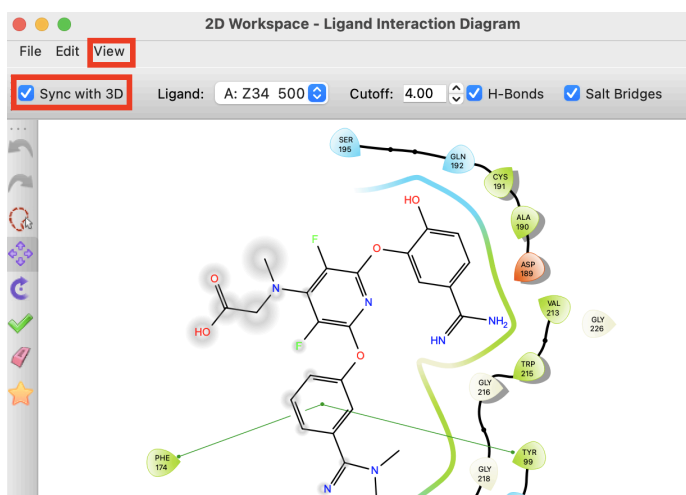
Generate a 2D interaction diagram



"Ligand Interaction" Diagram in the "Favorites toolbar".

In the **Favorites toolbar**, click **Ligand Interaction**

- The **2D Workspace - Ligand Interaction Diagram** opens



The "Ligand Interaction Diagram" with "Sync with 3D" turned on and LID legend open.

Check **Sync with 3D** and rotate the ligand in the Workspace

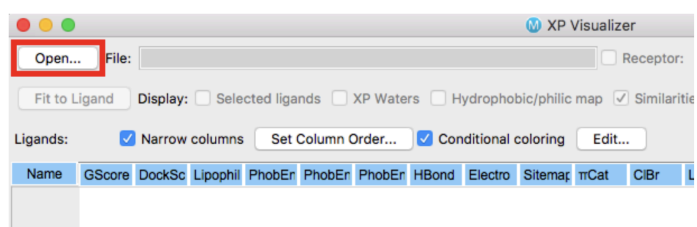
- Ligand orientation is changed in the 2D representation

Choose **View > LID Legend**

Note: Images can be saved via **File > Save Screenshot**

Note: The residue icon point indicates the direction of the sidechain

Visualize pre-docked XP results

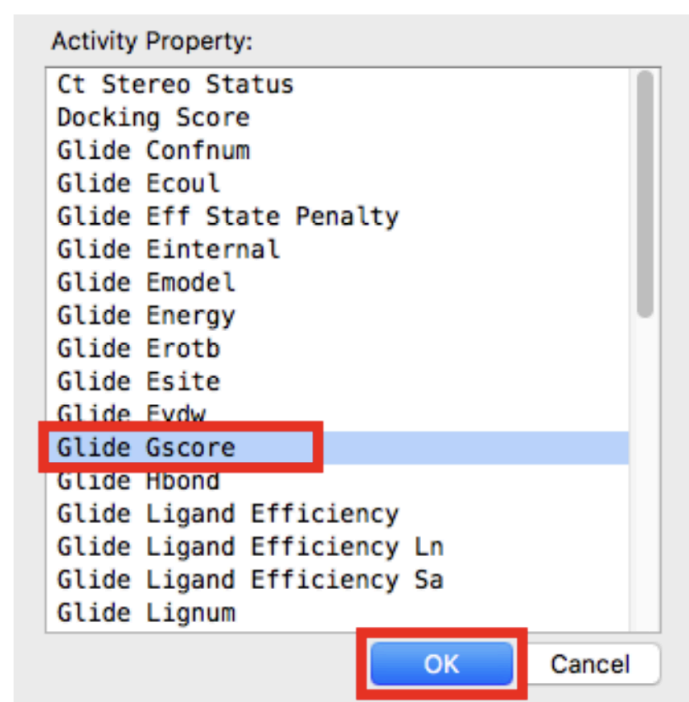


Go to **Tasks > Browse > Receptor-Based Virtual Screening > Visualize XP Interactions**

- XP Visualizer opens

Click **Open**

The XP Visualizer panel.



Choose **Glide Gscore** as the activity property, click **OK**

- The table is populated with the XP results
- Individual terms of the scoring function are colored as red (penalty) or blue (reward)

Choose the Activity Property.

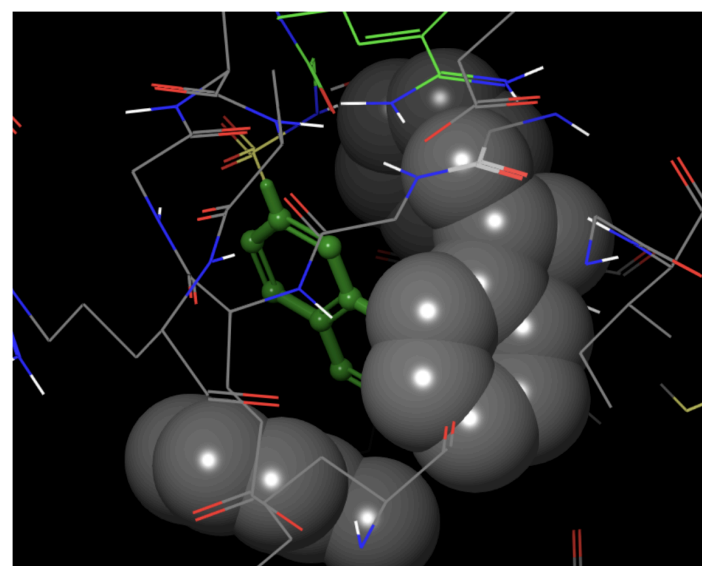
Ligands: (10 total, 10 shown, 1 selected) ☒ Narrow columns Set Column Order... ☒ Condi

Name	GScore	DockSc	Lipophil	PhobEr	PhobEr	PhobEr	HBond	Electro	Sitema	πCat	ClBr
16088	-11.9	-11.9	-5.5	-0.9	0.0	0.0	-2.9	-2.3	-0.3	0.0	0.0
15650	-11.1	-11.1	-7.0	-1.0	0.0	0.0	-0.9	-2.1	-0.1	0.0	0.0
1167	-10.2	-10.2	-4.8	-0.2	0.0	0.0	-2.5	-2.3	-0.3	0.0	0.0
15722	-10.0	-10.0	-5.4	-0.7	0.0	0.0	-1.8	-2.3	-0.1	0.0	0.0
689972	-9.9	-9.9	-5.2	-0.4	0.0	0.0	-2.4	-1.0	0.0	-1.9	0.0
612278	-7.1	-7.1	-5.3	-0.2	0.0	0.0	-1.0	-0.4	-0.1	0.0	0.0
494088	-6.5	-6.5	-3.3	-0.8	0.0	0.0	-1.3	-0.5	-0.4	0.0	0.0
334669	-6.1	-6.1	-4.8	0.0	0.0	0.0	-0.6	-0.3	-0.4	0.0	0.0

Show: All Selected Only **Export Data...** Export Structures... Reset Panel

Click **Export Data** to export the spreadsheet as a .csv file

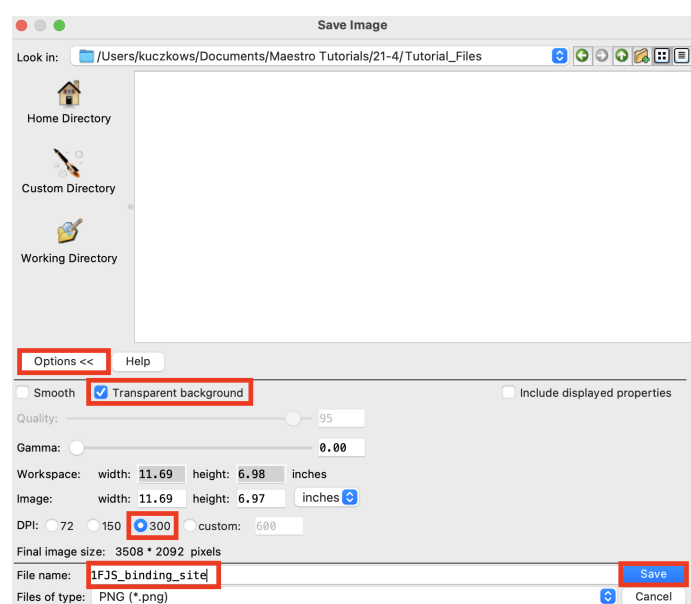
The XP Visualizer shows rewards (blue) and penalties (red) to the Glide Gscore.



Hydrophobic enclosure reward shown in the Workspace.

Click on the **indented colored entries** to visualize in the Workspace

Save an image of the Workspace



"Save Image" panel with expanded "Options" shown.

Go to **Workspace > Save Image As**
- The **Save Image** panel opens

Click **Options >>**

Check **Transparent background** and select **300** DPI

Change **File name** to **1FJS_binding_site**

Click **Save**
- A .png image of the Workspace is saved to your Working Directory

Note: If an item is highlighted in the Workspace, the image is saved with the selection highlights

Note: Go to **Tasks > Browse > Workspace and Project Table Operations** for more image options, including Ray Trace

Individual Exercise

Part A:

Perform the same preparation steps on the 2XIR crystal structure. What were some of the issues with this PDB structure from the first step of protein preparation?

Part B:

Search the Protein Data Bank for another protein-ligand complex and do the following:

- Import the .pdb file into Maestro
- Split the structure into Ligands, Waters, and Receptor
- Prepare the protein using the Protein Preparation Workflow
- Prepare the ligand using LigPrep (only cognate ligand)
- Detect the binding site using Receptor Grid Generation
- Dock all the conformations and tautomers generated from LigPrep back into the prepared protein using Schrödinger's Glide
- Take a screenshot of your docking results including the docking score and pose of the best docking pose.